Identifiability, optimal experimental design and closed-loop control for a stochastic model of meditation dynamics

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Abstract

Dynamical models that link default-mode-network (DMN) activity, information integration (Φ) and a reported "quality of consciousness" (QoC) during meditation are increasingly proposed, but three questions that precede any data analysis are rarely asked: whether the model is mathematically consistent, whether its parameters can be identified from the experience-sampling designs actually in use (a few self-report probes per session, sessions of 20–90 minutes, EEG features every few seconds), and what closed-loop control (neurofeedback) can achieve when the parameters are only estimated. We answer all three for a recently proposed three-variable DMN–Φ–QoC model. First, we prove that in the original formulation QoC is a high-pass filter of the derivatives of Φ and A_{DMN}, so its steady state is identically zero for every parameter value; the closed-form steady states printed with the model do not satisfy its own equations. We propose a linear-Gaussian state-space replacement (M1) with an Ornstein–Uhlenbeck DMN process and an algebraic read-out, and derive its stationary mean, covariance and Gaussian law in closed form. Second, using the exact expected Fisher information of a whole session we characterise structural identifiability for four observation configurations: ratings alone identify 6 of 10 parameters, ratings plus one EEG channel leave one exact scaling symmetry each (made explicit analytically), and ratings with both EEG channels identify all 12 parameters. Over a grid of 144 designs the Cramér–Rao bound (CRB) predicts the root-mean-square error of maximum-likelihood recovery within a factor of two in 24 of 24 tested cells (median ratio 0.98), which licenses the CRB as a design tool. The resulting design map shows that the DMN inhibition rate k_{inh} is the hardest parameter: it needs about eight subject-hours of two-channel EEG (e.g. 24 subjects $\times$ 20 min) for a relative error of 20 %, whereas A_0, α and β reach that level with 6 subjects $\times$ 20 min; uniform probe schedules are not beaten by the front-loaded, back-loaded or paired schedules we tested. Third, we recast the control problem as linear-quadratic-Gaussian (LQG) regulation. With true parameters LQG attains a session cost 14 times lower than bang-bang control and 38 times lower than no control, its closed-form stationary cost matches simulation within 3 %, and when the controller is designed on parameters

estimated from each design the median efficiency loss is a few per cent for designs that meet the identifiability criterion (3 % at the reference design) but reaches 38 % for the smallest design that fails it (6 subjects, 20-minute sessions). A global Sobol analysis over a $\times 0.25$–$\times 4$ parameter range shows that the identifiability of k_{inh} is governed by the DMN baseline and its coefficient of variation rather than by k_{inh} itself. All results are generated *in silico* from the model with a fixed seed and are fully reproducible; no claim is made about parameter values of real brains.

Keywords: meditation; default mode network; state-space model; Fisher information; structural identifiability; optimal experimental design; experience sampling; Kalman filter; LQG control; neurofeedback; Sobol sensitivity analysis.

1. Introduction

Meditation research has moved from describing mean differences between meditators and controls [6, 12] towards models of the *dynamics* of a session: the alternation between focused attention, mind wandering and meta-awareness [9, 17, 23], the regulation of the default mode network (DMN) [6, 14, 27], and theoretical accounts in terms of predictive processing and active inference [21, 33]. Several recent proposals go one step further and write explicit differential equations whose state variables are a DMN activity level, a measure of information integration Φ [24, 36] and a scalar "quality of consciousness" (QoC) that is supposed to be reported by the practitioner. One such model, published as part of a broader framework in Vu [40], is a three-variable ordinary differential equation (ODE) system which we denote M0 and reproduce in Section 2.

Models of this kind are attractive because they connect three streams of data that meditation studies routinely collect: a continuous electrophysiological signal (EEG features every few seconds), occasional self-reports obtained with experience-sampling probes [5, 17, 42], and an experimental manipulation (instruction or feedback). But before such a model can be fitted, three questions must be answered, and to our knowledge none of them has been addressed for any model of meditation dynamics:

(i) **Consistency.** Does the model have the qualitative behaviour it is claimed to have? In particular, does it possess a non-trivial steady state for the reported quantity?

(ii) **Identifiability and design.** Can the parameters be determined from the designs actually in use—N subjects, sessions of 20–90 minutes, probes every 0.5–4 minutes, EEG of a given signal-to-noise ratio—and if so, how many subjects and how long a session are needed for a given precision?

(iii) **Control.** If the model is used to drive a closed loop (neurofeedback, 30, 34), what does the loop gain over simple strategies, and how much of that gain

survives when the controller is designed on estimated rather than true parameters?

Questions (ii) and (iii) are classical in systems identification and systems biology [4, 8, 20, 22, 29, 31, 38, 41, 43], and (iii) in stochastic control [1, 2], but their combination has not been applied to the experience-sampling setting, where a latent two-dimensional process is observed through a dense noisy channel (EEG) and a sparse noisy channel (ratings), and where the design variables are the number of subjects, the session length and the probe interval.

Contributions.

1. **A correction of the original model** (Section 2, Section 4.1). We prove that in M0 the QoC equation is a first-order high-pass filter acting on $\alpha\dot{\Phi} - \beta\dot{A}$, so that $\mathrm{QoC}^* = 0$ for all parameter values and all constant drives, and that the steady-state formulas printed with the model violate the model's own equations (Propositions 2.1–2.3). We propose a linear-Gaussian state-space model M1 in which the DMN level is an Ornstein–Uhlenbeck process around a baseline A_0 and QoC is an algebraic read-out, and give its stationary mean, covariance and Gaussian law in closed form (Propositions 2.4–2.5).
2. **Identifiability and a quantitative design map** (Sections 4.2 and 4.4). Because M1 is linear-Gaussian, the joint distribution of all observations of a session is Gaussian with closed-form moments, and the expected Fisher information can be computed *exactly*, without simulation. We use it (a) to establish structural identifiability by Rothenberg's rank criterion for four observation configurations, exhibiting the exact symmetries that make single-EEG-channel designs non-identifiable; (b) to compute Cramér–Rao bounds (CRB) over a grid of 144 designs; (c) to validate the CRB against maximum-likelihood (ML) recovery on simulated data at 24 designs; and (d) to derive minimal designs and simple scaling rules for each parameter.
3. **Closed-loop control and its dependence on design quality** (Section 4.5). We formulate the control problem as linear-quadratic-Gaussian (LQG) regulation of QoC towards a target, compare it with a bang-bang policy using the same state estimator, give the closed-loop stationary cost in closed form, and quantify the loss of efficiency when the controller is designed on parameter estimates obtained from each experimental design. This links the design map directly to the achievable performance of a model-based neurofeedback loop.
4. **Global sensitivity** (Section 4.6). A variance-based Sobol analysis over a $\times 0.25$–$\times 4$ range around the nominal parameters shows which conclusions are robust and which parameters govern identifiability and control performance.

What this paper does not claim. All parameter values are nominal working points; we make no statement about the parameters of real brains, nor about whether meditation "increases" Φ or QoC. The contribution is methodological: a complete, reproducible analysis of what a specific class of models can and cannot deliver *before* data are collected, together with design recommendations that experimenters can apply directly. All results are produced by a single command from the accompanying code with a fixed random seed (Section 3.10).

2. Models

2.1. The original model M0

The model of Vu [40, eqs. 52–54] describes a session by three scalar variables: the DMN activity $A(t) \in [0, A_{\max}]$, an integration measure $\Phi(t) \geq 0$ and the quality of consciousness $\mathrm{QoC}(t)$:

$$\dot{A} = -k_{\mathrm{inh}}\, A + k_{\mathrm{spont}}\, \xi(t), \tag{1}$$

$$\dot{\Phi} = k_{\mathrm{int}}\Big(1 - \frac{A}{A_{\max}}\Big) - k_d\, \Phi, \tag{2}$$

$$\dot{\mathrm{QoC}} = \alpha\, \dot{\Phi} - \beta\, \dot{A} - \gamma\, \mathrm{QoC}. \tag{3}$$

Here k_{inh} is the rate at which practice suppresses DMN activity, $\xi(t)$ a spontaneous activation drive treated as a zero-mean Gaussian process (eq. 51 of 40) of standard deviation σ_ξ, k_{int} and k_d the integration and decay rates of Φ, $\alpha, \beta > 0$ gains and $\gamma > 0$ a relaxation rate. The point values printed with the model (Table 5 of 40) are $k_{\mathrm{inh}} = 0.043$, $k_{\mathrm{spont}} = 0.027$, $k_{\mathrm{int}} = 0.081$, $k_d = 0.034\ \mathrm{min}^{-1}$, $\alpha = 2.34$, $\beta = 1.87$, $\gamma = 0.19\ \mathrm{min}^{-1}$; no values are given for $A_{\max}$ and σ_ξ, and we take $A_{\max} = 1$ (the normalised DMN activity used in that paper's figures) and $\sigma_\xi = 0.05$. These numbers are used here only to evaluate the printed formulas; none of the estimates, confidence intervals or sample sizes of that table is used as evidence anywhere in this paper. The following closed-form steady states are stated, "setting $\dot{A}_{\mathrm{DMN}} = \dot{\Phi} = \dot{\mathrm{QoC}} = 0$" [40, eqs. 55–57]:

$$\begin{aligned} A^*_{\mathrm{p}} &= \frac{k_{\mathrm{spont}}}{k_{\mathrm{inh}}}\sqrt{2\pi\sigma_\xi^2}, \qquad \Phi^*_{\mathrm{p}} = \frac{k_{\mathrm{int}}}{k_d}\Big(1 - \frac{A^*_{\mathrm{p}}}{A_{\max}}\Big), \\ \mathrm{QoC}^*_{\mathrm{p}} &= \frac{1}{\gamma}\left(\alpha k_{\mathrm{int}}\Big(\frac{A_{\max} - A^*_{\mathrm{p}}}{A_{\max}\, k_d}\Big) + \beta\, \frac{k_{\mathrm{spont}}}{k_{\mathrm{inh}}}\sqrt{2\pi\sigma_\xi^2}\right). \end{aligned} \tag{4}$$

(The subscript p marks the printed values; the last term of $\mathrm{QoC}^*_{\mathrm{p}}$ equals βA^*_{p}.) With the printed parameters, (4) gives $A^*_{\mathrm{p}} = 0.079$, $\Phi^*_{\mathrm{p}} = 2.19$ and $\mathrm{QoC}^*_{\mathrm{p}} = 27.8$ (Table 2). We now show that (4) cannot be a steady state of (1)–(3).

2.2. What is wrong with M0

Let $v(t) = \alpha\Phi(t) - \beta A(t)$ denote the *drive* of the QoC equation. Then (3) reads $\dot{\text{QoC}} = \dot{v} - \gamma\text{QoC}$, a first-order linear equation in QoC forced by the derivative of v.

Proposition 2.1 (Steady states of M0; P1)**.** *For any constant drive $\xi(t) \equiv \bar{\xi}$ (in particular $\bar{\xi} = 0$, the mean of the noise), the unique equilibrium of (1)–(3) is*

$$A^* = \frac{k_{\text{spont}}\bar{\xi}}{k_{\text{inh}}}, \qquad \Phi^* = \frac{k_{\text{int}}}{k_d}\Big(1 - \frac{A^*}{A_{\max}}\Big), \qquad \text{QoC}^* = 0, \tag{5}$$

for every admissible value of $(k_{\text{inh}}, k_{\text{spont}}, k_{\text{int}}, k_d, \alpha, \beta, \gamma, A_{\max})$. With zero-mean noise, $A^* = 0$, $\Phi^* = k_{\text{int}}/k_d$.

Proposition 2.2 (QoC is a high-pass filter of the drive; P2)**.** *For any absolutely continuous drive v and any $\gamma > 0$,*

$$\text{QoC}(t) = e^{-\gamma t}\big[\text{QoC}(0) - v(0)\big] + v(t) - \gamma\int_0^t e^{-\gamma(t-s)}\, v(s)\,\mathrm{d}s. \tag{6}$$

Equivalently, the transfer function from v to QoC is $H(s) = s/(s+\gamma)$, whose gain at zero frequency is $H(0) = 0$ and whose modulus is $|H(j\omega)| = \omega/\sqrt{\omega^2+\gamma^2}$. In particular, if $v(t) \to v_\infty$ then $\text{QoC}(t) \to 0$.

Proposition 2.3 (The printed steady states violate the model; P3)**.** *The triple (4) is not an equilibrium of (1)–(3) for any value of $\bar{\xi}$: at any point where $\dot{A} = \dot{\Phi} = 0$ the right-hand side of (3) equals $-\gamma\text{QoC}$, which is $-\gamma\text{QoC}^*_{\text{p}} \neq 0$ at (4). With the printed parameters and $\bar{\xi} = 0$ the residual vector $(\dot{A}, \dot{\Phi}, \dot{\text{QoC}})$ at (4) is $(-3.4\times10^{-3},\ 0,\ -5.28)$, of norm 5.28.*

Proofs are given in Appendix A. The first component of (4) is also inconsistent with the stated noise model: eq. 55 makes A^* proportional to $\sqrt{2\pi\sigma^2_{\bar{\xi}}}$, whereas for the zero-mean drive of eq. 51 the equilibrium of (1) is $A^* = 0$ (Proposition 2.1); a positive A^* would require a positive mean drive $\bar{\xi}$, which is not part of the model. The consequence of Proposition 2.2 is qualitative, not numerical: in M0 the reported quantity is a *transient*. Any constant level of DMN activity and integration, however favourable, produces zero QoC after a few multiples of $1/\gamma$ (Figure 1a). A model intended to describe a sustained state of consciousness therefore needs QoC to depend on the *levels* of its inputs, not only on their rates of change. The origin of the error in (4) is visible in its third component: since $\alpha k_{\text{int}}(A_{\max} - A^*_{\text{p}})/(A_{\max}k_d) = \alpha\Phi^*_{\text{p}}$, eq. 57 reads $\text{QoC}^*_{\text{p}} = (\alpha\Phi^*_{\text{p}} + \beta A^*_{\text{p}})/\gamma$, i.e. (3) was solved with the *levels* substituted for the derivatives $\dot{\Phi}$, $\dot{A}$ (and with the sign of the β term reversed); the division by γ then inflates QoC^* by an order of magnitude (Table 2: $(2.34\times2.195 + 1.87\times0.079)/0.19 = 27.8$). Notably, the original framework itself does not use (3) when it needs a value of QoC: its Algorithm 2 (line 21) and its control objective (eq. 69) evaluate QoC as the algebraic read-out $\alpha\Phi - \beta A_{\text{DMN}} + \gamma_0$ [40]. The model proposed next adopts exactly this read-out and thereby removes the inconsistency between the equations and the algorithm.

2.3. The proposed model M1

We keep the two mechanistic ingredients of M0—a DMN level that is suppressed by practice and an integration variable that is fed by DMN disengagement—and repair the two defects identified above: the DMN process is given a non-zero baseline A_0 so that it has a proper stationary regime, and QoC becomes the static read-out of the state already used in Algorithm 2 of Vu [40], so that it inherits a non-zero stationary level. (The symbol A_0 denotes a baseline here, not the initial amplitude of eq. 51 of 40.) Rates are per minute.

$$\mathrm{d}A = -k_{\mathrm{inh}}\,(A - A_0)\,\mathrm{d}t + \sigma_A\,\mathrm{d}W_A, \tag{7}$$

$$\mathrm{d}\Phi = \Big[k_{\mathrm{int}}\Big(1 - \frac{A}{A_{\max}}\Big) - k_d\,\Phi\Big]\,\mathrm{d}t + \sigma_\Phi\,\mathrm{d}W_\Phi, \tag{8}$$

$$\mathrm{QoC} = \alpha\,\Phi - \beta\,A + \gamma_0, \tag{9}$$

with independent Wiener processes W_A, W_Φ. In state-space form, $\mathbf{x} = (A, \Phi)^\top$ obeys the linear stochastic differential equation $\mathrm{d}\mathbf{x} = (M\mathbf{x} + \mathbf{u})\,\mathrm{d}t + G\,\mathrm{d}\mathbf{W}$ with

$$M = \begin{pmatrix} -k_{\mathrm{inh}} & 0 \\ -k_{\mathrm{int}}/A_{\max} & -k_d \end{pmatrix}, \qquad \mathbf{u} = \begin{pmatrix} k_{\mathrm{inh}}A_0 \\ k_{\mathrm{int}} \end{pmatrix}, \qquad G = \mathrm{diag}(\sigma_A, \sigma_\Phi),$$
$$\mathrm{QoC} = \mathbf{h}^\top\mathbf{x} + \gamma_0, \qquad \mathbf{h} = \begin{pmatrix} -\beta \\ \alpha \end{pmatrix}. \tag{10}$$

M is lower-triangular with eigenvalues $-k_{\mathrm{inh}}, -k_d < 0$, so the process is stable and has a unique stationary distribution.

Proposition 2.4 (Stationary moments of M1; P4)**.** *The stationary mean is* $\boldsymbol{\mu} = -M^{-1}\mathbf{u}$*, i.e.*

$$A^* = A_0, \qquad \Phi^* = \frac{k_{\mathrm{int}}}{k_d}\Big(1 - \frac{A_0}{A_{\max}}\Big), \qquad \mathrm{QoC}^* = \alpha\Phi^* - \beta A_0 + \gamma_0, \tag{11}$$

and the stationary covariance P is the unique solution of the Lyapunov equation $MP + PM^\top + GG^\top = 0$,

$$P_{AA} = \frac{\sigma_A^2}{2k_{\mathrm{inh}}}, \qquad P_{A\Phi} = -\frac{k_{\mathrm{int}}}{A_{\max}}\,\frac{P_{AA}}{k_{\mathrm{inh}} + k_d}, \qquad P_{\Phi\Phi} = \frac{\sigma_\Phi^2 - 2k_{\mathrm{int}}P_{A\Phi}/A_{\max}}{2k_d}. \tag{12}$$

QoC^* *is non-zero for generic parameters.*

Proposition 2.5 (Stationary law of QoC; P5)**.** *In the stationary regime $\mathbf{x}(t)$ is a Gaussian process with mean $\boldsymbol{\mu}$, covariance P and autocovariance $\mathrm{Cov}(\mathbf{x}(t+\tau), \mathbf{x}(t)) = e^{M\tau}P$ for $\tau \geq 0$. Consequently $\mathrm{QoC}(t) \sim \mathcal{N}(\mathrm{QoC}^*, \mathbf{h}^\top P\mathbf{h})$ with autocovariance $\mathbf{h}^\top e^{M\tau}P\mathbf{h}$.*

2.4. Observation model and nominal parameters

A session of length T minutes is divided into epochs of $\Delta = 10$ s. In every epoch an EEG feature vector is extracted and modelled as a noisy direct observation of the

latent state,

$$\mathbf{y}_t = \mathbf{x}_t + \boldsymbol{\eta}_t, \qquad \boldsymbol{\eta}_t \sim \mathcal{N}\big(0, \mathrm{diag}(r_A^2, r_\Phi^2)\big), \tag{13}$$

and at the probe epochs $t \in \mathcal{P}$ (every T_p minutes) the practitioner reports a rating,

$$\mathrm{rating}_t = \mathrm{QoC}_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, r_Q^2). \tag{14}$$

This mirrors the structure of existing experience-sampling meditation data sets such as ds001787 [5], in which EEG is recorded continuously and probes occur every few minutes. The DMN channel y_A would in practice be an EEG proxy of DMN activity (e.g. posterior alpha or a source-localised feature) and y_Φ a complexity or integration measure; the mapping of real features to (A, Φ) is outside the scope of this paper, which asks what could be learned *if* such features existed with noise levels r_A, r_Φ. Four observation configurations are considered: (a) ratings only, (b) ratings $+\, y_A$, (c) ratings $+\, y_\Phi$, (d) ratings $+\, y_A + y_\Phi$.

The free parameter vector is

$$\boldsymbol{\theta} = (k_{\mathrm{inh}}, A_0, \sigma_A, k_{\mathrm{int}}, k_d, \sigma_\Phi, \alpha, \beta, \gamma_0, r_A, r_\Phi, r_Q) \in \mathbb{R}^{12}, \tag{15}$$

with $A_{\max} = 1$ fixed (it sets the unit of A). Nominal values (Table 1) are a working point, not estimates: $k_{\mathrm{int}}, k_d, \alpha, \beta$ are the point values printed with M0; $k_{\mathrm{inh}} = 0.2$ per minute (DMN time constant 5 min) is deliberately larger than the printed 0.043 per minute (time constant 23 min), because with the latter a session of 20 to 45 minutes would span only one or two DMN time constants and the model would be close to non-stationary over the session; the integration time constant is $1/k_d \approx 29$ min; $A_0 = 0.6$ places the baseline inside $(0, A_{\max})$; the noise intensities give stationary standard deviations $\mathrm{sd}(A) = 0.158$, $\mathrm{sd}(\Phi) = 0.240$ and $\mathrm{sd}(\mathrm{QoC}) = 0.691$ (see Table 1). Every conclusion is re-examined on a $\times 0.25$–$\times 4$ range around this point in Section 4.6, whose lower end for k_{inh} (0.05 min^{-1}) is close to the printed value, and Section 5.1 discusses how the design map should be read for slower DMN dynamics.

3. Methods

3.1. Exact discretisation

Sampling (10) at the epoch length Δ gives the exact linear-Gaussian recursion

$$\begin{aligned} \mathbf{x}_{k+1} &= F\mathbf{x}_k + \mathbf{c} + \mathbf{w}_k, \qquad \mathbf{w}_k \sim \mathcal{N}(0, Q), \\ F = e^{M\Delta}, \qquad \mathbf{c} &= M^{-1}(F - I)\mathbf{u}, \qquad Q = \int_0^\Delta e^{Ms} G G^\top e^{M^\top s}\, \mathrm{d}s, \end{aligned} \tag{16}$$

where Q is computed with Van Loan's matrix-exponential construction [37] (Appendix C). No Euler approximation is used anywhere; all simulations, likelihoods and Fisher matrices are exact for the continuous-time model. The initial state of

Table 1: Nominal parameters of M1 (rates per minute) and the derived stationary quantities.

Parameter	Meaning	Value	Derived quantity
k_{inh}	DMN inhibition (relaxation) rate	0.20	$1/k_{\text{inh}} = 5$ min
A_0	DMN baseline	0.60	$A^* = 0.60$
σ_A	DMN noise intensity	0.10	$\text{sd}_\infty(A) = 0.158$, $\text{CV}_\infty(A) = 0.26$
k_{int}	Φ integration rate	0.081	
k_d	Φ decay rate	0.034	$1/k_d \approx 29$ min, $\Phi^* = 0.953$
σ_Φ	Φ noise intensity	0.05	$\text{sd}_\infty(\Phi) = 0.240$
α	gain $\Phi \to$ QoC	2.34	
β	gain $A \to$ QoC	1.87	$\text{QoC}^* = 1.108$, $\text{sd}_\infty(\text{QoC}) = 0.691$
γ_0	rating offset	0	(0.3 in the structural analysis)
r_A, r_Φ	EEG observation noise sd	0.10	
r_Q	rating noise sd	0.30	
$A_{\max}$	DMN ceiling (fixed)	1	
Δ	epoch length	10 s	

every simulated session is drawn from the stationary law $\mathcal{N}(\boldsymbol{\mu}, P)$, which is also the prior used by the Kalman filter and the Fisher computation.

3.2. Likelihood and maximum-likelihood estimation

For one session the log-likelihood of $\boldsymbol{\theta}$ is evaluated by the Kalman filter [11, 16, 19] with sequential scalar updates (one update per observed channel per epoch; Appendix D), which handles the irregular rating channel naturally. Sessions of different subjects are independent, so N subjects contribute additively. Positive parameters $(k_{\text{inh}}, \sigma_A, k_{\text{int}}, k_d, \sigma_\Phi, r_A, r_\Phi, r_Q)$ are estimated on the log scale, $A_0, \alpha, \beta, \gamma_0$ on the linear scale; we write $\boldsymbol{\theta}$ for this transformed vector throughout. Maximisation uses L-BFGS-B [7] with two starting points per fit: the first is the true parameter vector with every positive parameter multiplied by an independent $\exp\big(\mathcal{N}(0, 0.3^2)\big)$ factor (so that the optimiser never starts at the truth), the second adds $\mathcal{N}(0, 0.3^2)$ noise to every transformed coordinate of the first; the better of the two optima is retained. The Kalman log-likelihood was verified against the direct Gaussian log-likelihood of the stacked observation vector (Section 3.3) to 10^{-11}.

3.3. Exact expected Fisher information

Because M1 is linear-Gaussian and stationary, *all* observations of a session form a jointly Gaussian vector $\mathbf{Y} \sim \mathcal{N}(\mathbf{m}(\boldsymbol{\theta}), \Sigma(\boldsymbol{\theta}))$ with closed-form moments: with $\text{Cov}(\mathbf{x}_t, \mathbf{x}_s) = F^{t-s}P$ for $t \geq s$, the covariance of the stacked latent path is a block-Toeplitz matrix, and each observation is a fixed linear functional of one $\mathbf{x}_t$ plus independent noise (Appendix E). The expected Fisher information matrix for one session is therefore available exactly,

$$\mathcal{I}_{ij}(\boldsymbol{\theta}) = \frac{\partial \mathbf{m}^\top}{\partial \theta_i} \Sigma^{-1} \frac{\partial \mathbf{m}}{\partial \theta_j} + \frac{1}{2} \operatorname{tr}\Big(\Sigma^{-1} \frac{\partial \Sigma}{\partial \theta_i} \Sigma^{-1} \frac{\partial \Sigma}{\partial \theta_j} \Big), \tag{17}$$

and N identically designed subjects give $N\mathcal{I}$. Derivatives of $(\mathbf{m}, \Sigma)$ are central finite differences in the transformed parameters. No Monte Carlo noise enters any bound reported below. The implementation was checked against the Kalman likelihood (agreement 10^{-13}) and against the empirical covariance of Kalman score vectors over 3,000 simulated sessions (agreement within 5 %).

3.4. Structural identifiability

By Rothenberg's theorem [31], in a regular model a parameter point is locally identifiable if and only if the information matrix is non-singular there; the null space of $\mathcal{I}$ consists of the parameter directions along which the distribution of the data does not change to first order. We compute $\mathcal{I}$ for a long, densely probed session (90 min, probes every minute) in each configuration (a)–(d), with $\gamma_0 = 0.3$ so that no symmetry is masked by a zero value, normalise it to unit diagonal, and declare the directions whose relative eigenvalue is below 10^{-7} non-identifiable. Parameters that do not enter the likelihood at all in a configuration (e.g. r_Φ when y_Φ is not recorded) are reported as *absent*. For each null direction one parameter is fixed ("normalised") at its nominal value, and the CRB of the remaining parameters is computed from the reduced information matrix. The numerically found null directions are compared with the exact symmetries derived in Appendix B.

3.5. Practical identifiability

For an identifiable configuration the Cramér–Rao bound [10, 28] on the covariance of any unbiased estimator is $(N\mathcal{I})^{-1}$. We report the *relative* CRB: the square root of the diagonal element for log-scale parameters (a relative standard error to first order), and the square root divided by $|\theta|$ for linear-scale parameters; for $\gamma_0 = 0$ the absolute standard error is given. All 12 parameters are always estimated jointly, so the bound accounts for all correlations among them. Profile likelihoods [29] are computed for the three rate constants on one simulated data set of the reference design: each parameter is moved over nine points spanning ± 0.8 log units around its true value while all other parameters are re-optimised.

3.6. Design grid and validation of the bound

The *reference design*, patterned on ds001787, is $N = 12$ subjects, $T = 45$ min, $T_p = 2$ min, $r_A = r_\Phi = 0.1$, configuration (d). The design grid crosses $N \in \{6, 12, 24, 48\}$, $T \in \{20, 45, 90\}$ min, $T_p \in \{0.5, 1, 2, 4\}$ min and EEG noise $r = r_A = r_\Phi \in \{0.05, 0.1, 0.2\}$, i.e. 144 cells, in configuration (d); the CRB is computed in every cell. A design is called *adequate* for a parameter when its relative CRB is at most 20 %.

A bound is useful only if it predicts the error actually achieved. In 24 representative cells spanning the grid, five independent data sets were simulated per cell (seeds a deterministic function of the cell and replicate index), each fitted by ML as

in Section 3.2, and the root-mean-square error (RMSE) over the five fits computed *in the same scale as the bound* (log for positive parameters, relative for linear ones). Gate M1 of the analysis is passed when, in at least 80 % of the cells, at least half of the four key parameters $(k_{\text{inh}}, A_0, \alpha, \beta)$ have RMSE/CRB $\in [1/2, 2]$. The 24 cells cost about seven hours on a four-core machine and are the only computationally expensive part of the study.

3.7. Probe schedules

Holding the number of probes fixed (22 probes in a 45-min session), we ask whether a non-uniform schedule reduces the Ds-optimality criterion [3, 26] $\log\det\big[(\mathcal{I}^{-1})_{\mathcal{K}\mathcal{K}}\big]$, the log-volume of the confidence ellipsoid of the key parameters $\mathcal{K} = (k_{\text{inh}}, A_0, \alpha, \beta)$ with all 12 parameters estimated. Two interpretable families are compared with the uniform schedule: power spacing $t_k = T\,((k+\frac{1}{2})/n)^p$ with $p \in [0.4, 2.5]$ (15 values; $p < 1$ front-loads, $p > 1$ back-loads, $p = 1$ is uniform) and paired probes ($n/2$ pairs, partner $g \in \{1, 2, 3, 6, 12\}$ epochs later, which inform fast time constants). Efficiency relative to uniform is $\exp\big[-(\log\det_{\text{opt}} - \log\det_{\text{unif}})/|\mathcal{K}|\big]$, the ratio of geometric-mean standard errors.

3.8. Closed-loop control

Controlled model. An intervention $u(t)$ (instruction, feedback, stimulation) is assumed to act on the DMN equation, $\mathrm{d}A = [-k_{\text{inh}}(A - A_0) - b\,u]\,\mathrm{d}t + \sigma_A\,\mathrm{d}W_A$ with $b = 1$, bounded by $|u| \le u_{\max} = 2$. Exact zero-order-hold discretisation gives the input matrix $B = M^{-1}(F - I)\mathbf{b}$, $\mathbf{b} = (-b, 0)^\top$. The controller observes the EEG channels (13) every epoch; ratings are not used for control.

LQG regulator. The target $\text{QoC}_{\text{tgt}} = 2.0$ (about 1.3 stationary standard deviations above QoC^*) is reached at the unique equilibrium $(\mathbf{x}_{\text{ss}}, u_{\text{ss}})$ with $F\mathbf{x}_{\text{ss}} + \mathbf{c} + Bu_{\text{ss}} = \mathbf{x}_{\text{ss}}$ and $\mathbf{h}^\top\mathbf{x}_{\text{ss}} + \gamma_0 = \text{QoC}_{\text{tgt}}$. On the deviations the quadratic cost $\sum_t[(\text{QoC}_t - \text{QoC}_{\text{tgt}})^2 + \lambda u_t^2]$, $\lambda = 0.1$, is minimised by the certainty-equivalence LQG controller [1, 2]: $u_t = u_{\text{ss}} - K(\hat{\mathbf{x}}_t - \mathbf{x}_{\text{ss}})$ with the discrete-time LQR gain K (algebraic Riccati equation with state weight $\mathbf{h}\mathbf{h}^\top$) and $\hat{\mathbf{x}}_t$ from the stationary Kalman filter for $\mathbf{y}_t = \mathbf{x}_t + \boldsymbol{\eta}_t$; u_t is clipped to $[-u_{\max}, u_{\max}]$. For the nominal model $K = (-3.61,\ 4.60)$, $\mathbf{x}_{\text{ss}} = (0.480, 1.238)$ and $u_{\text{ss}} = 0.024$.

Relation to the original control formulation. Vu [40] states the control task in two ways: Algorithm 2 (line 5) applies proportional feedback $u_{\text{stim}} \leftarrow K\,(\text{QoC}_{\text{target}} - \text{QoC}(t))$ towards a target, while Sec. 9.4 and 10.1 (eqs. 69–75) maximise the terminal or discounted value of QoC subject to the ODE, a bound $\|u_t\| \le \epsilon$ and non-negativity, and conclude from Pontryagin's principle that the optimal stimulation is bang-bang, $u^* = \epsilon\,\text{sign}(\lambda_A)$. Two points in that formulation need repair before any controller can be derived. First, the control variable does not appear in the constraint dynamics

(eqs. 70–71 are the uncontrolled ODE), so the stated problem has no channel through which u acts; the controlled model above supplies one by letting u enter the DMN equation with gain b. Second, the problem is called linear-quadratic although its objective (eq. 69) is linear in the state and the input is unpenalised; for such a problem the Pontryagin solution is indeed bang-bang, but as soon as the objective is quadratic (hold QoC near a target at moderate effort), the optimal policy is linear feedback on the state estimate, i.e. LQG. The LQG formulation above is therefore the stochastic, state-estimated version of the target-tracking reading of Algorithm 2 with the feedback gain chosen optimally rather than set by hand, whereas the bang-bang law is the solution of the linear-objective reading.

Comparators. *Bang-bang*: $u_t = \pm u_{\max}$ according to the sign of $\mathrm{QoC}_{\mathrm{tgt}} - (\mathbf{h}^\top \hat{\mathbf{x}}_t + \gamma_0)$, using the *same* Kalman estimate, so that the comparison isolates the control law from the estimator. *No control*: $u \equiv 0$. Because the bang-bang law solves a different problem, we report, besides the quadratic cost, the two neutral components: tracking mean-squared error $\overline{(\mathrm{QoC}_t - \mathrm{QoC}_{\mathrm{tgt}})^2}$ and the root-mean-square input. All policies are run on the true model over 270 epochs (45 min) with common random numbers, 20 trials.

Closed-form stationary cost. For the unclipped LQG loop the augmented state $(\mathbf{x}_t, \hat{\mathbf{x}}_t)$ is a linear Gaussian recursion whose stationary mean and covariance follow from a 4×4 discrete Lyapunov equation (Appendix F); the stationary cost is then explicit. Gate M2 requires that LQG with true parameters beats both comparators and that this closed form agrees within 20 % with a long simulation (4,000 epochs, 1,000 discarded). The closed form is what the sensitivity analysis evaluates.

Robustness to estimation error. For each of the 24 validated design cells and each of its five ML fits, an LQG controller (gain *and* filter) is designed on the *estimated* parameters and run on the true model of that cell (nominal parameters with the cell's EEG noise level) for 10 trials; the efficiency loss is $(\mathrm{cost}_{\mathrm{est}} - \mathrm{cost}_{\mathrm{true}})/\mathrm{cost}_{\mathrm{true}}$. We report the median and 90th percentile over fits.

3.9. Global sensitivity analysis

First-order (S_1, Saltelli estimator) and total-order (S_T, Jansen estimator) Sobol indices [18, 32, 35] are computed for eight inputs, $(k_{\mathrm{inh}}, A_0, \mathrm{CV}_A, k_{\mathrm{int}}, k_d, \sigma_\Phi, \alpha, \beta)$, each log-uniform on $[\frac{1}{4}, 4] \times$nominal, except that the DMN noise is parametrised by its stationary coefficient of variation $\mathrm{CV}_A = \sigma_A/(A_0\sqrt{2k_{\mathrm{inh}}}) \in [0.05, 0.4]$ rather than by σ_A: varying σ_A independently drives A below zero for a substantial fraction of the time at the corners of the box (30 % at $\sigma_A \times 4$, $k_{\mathrm{inh}} \times \frac{1}{4}$), whereas $\mathrm{CV}_A \leq 0.4$ keeps $\Pr(A < 0) \leq 0.6\,\%$. Four deterministic outputs are analysed: QoC^*; $\mathrm{Var}_\infty(\mathrm{QoC}) = \mathbf{h}^\top P \mathbf{h}$; the relative CRB of k_{inh} at the reference design; and the LQG efficiency, the ratio of the closed-form stationary LQG cost to the cost without control. The last

Table 2: Steady states of the DMN–Φ–QoC model with the parameters printed in Vu [40]. "M0, corrected" is the equilibrium (5) with zero-mean noise; "M1" is (11) at the nominal parameters of Table 1.

	Printed, eqs. (55)–(57) of [40]	M0, corrected (Prop. 2.1)	M1 (Prop. 2.4)
A^*	0.0787	0	0.600
Φ^*	2.195	2.382	0.953
QoC^*	27.81	0	1.108
$\mathrm{Var}_\infty(\mathrm{QoC})$	–	–	0.477

three are positive and vary over orders of magnitude across the box, so they are analysed on the log scale (on the raw scale a handful of extreme samples dominate the variance, indices clip at 1 and $S_1 \ll S_T$ spuriously); QoC^* can change sign and stays on the raw scale. The Saltelli matrices are built from a scrambled Sobol sequence [25] with base size $n = 256$, i.e. $n(8+2) = 2{,}560$ model evaluations. Ninety-five per cent confidence intervals are obtained by bootstrapping the rows ($B = 1{,}000$), and convergence is checked by recomputing S_T from the first half of the sample.

3.10. Reproducibility and verification gates

Every setting above lives in one configuration module; a snapshot is written next to every result. The analysis is organised as five packages with three machine-checked gates: M0 (Propositions 2.1–2.5 verified numerically on the actual ODE and SDE, not on the formulas: equilibrium and Lyapunov residuals below 10^{-6}, DC attenuation of the step response above 10^4, sinusoidal amplitude errors below 1 %, Kolmogorov–Smirnov $p > 0.01$ on thinned samples), M1 (CRB predicts ML error, Section 3.6) and M2 (control, Section 3.8). All numbers, tables and figures in this paper are produced from seed 42 by

```
python main.py model all --seed 42 --n-base 256 --profiles
python main.py gates
```

followed by the figure script; 85 unit tests cover the model, filter, Fisher matrix, gates and figure pipeline. Computations use NumPy and SciPy [15, 39]. Code and outputs will be archived with a DOI at publication.

4. Results

4.1. The original model has no sustained QoC; the proposed model does

Table 2 and Figure 1 summarise Propositions 2.1–2.3. The equilibrium of M0 under zero-mean noise has $\mathrm{QoC}^* = 0$ exactly, and the step response in Figure 1a shows that this is not an artefact of the particular equilibrium: a sustained change of DMN activity and integration produces a sustained change of the *drive* but no sustained QoC. The printed closed forms differ from the true equilibrium in all three

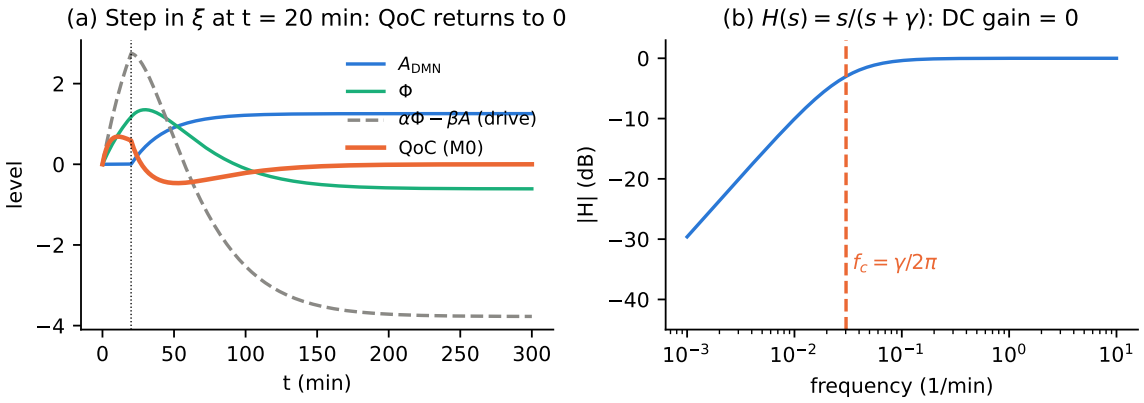


Figure 1: Proposition 2.2 on the actual ODE (1)–(3) with the printed parameters. (a) A step in the drive ξ at $t = 20$ min moves A and Φ to new levels and the drive $\alpha\Phi - \beta A$ to -3.77; QoC responds transiently and returns to zero (final value -8×10^{-9}, attenuation $> 10^8$). (b) Bode magnitude of $H(s) = s/(s+\gamma)$: zero gain at DC, corner frequency $\gamma/2\pi$. Sinusoidal drives at $\omega = 0.05, 0.2, 1.0$ rad/min reproduce $|H(j\omega)|$ with relative errors 0.14 %, 0, 0.

components and their residual in the model equations is of order $\gamma \mathrm{QoC}_{\mathrm{p}}^{*} \approx 5.3$ per minute, two orders of magnitude larger than the largest rate in the model.

M1 removes the defect (Figure 2). The Lyapunov residual of (12) is zero to machine precision, the stationary level is $\mathrm{QoC}^{*} = 1.108$ with variance 0.477, and the empirical stationary law of a long simulation is indistinguishable from the predicted Gaussian. Pooling several independent chains for the Kolmogorov–Smirnov test is deliberate: a single path of an autocorrelated process can sit two to three standard errors from its mean for hundreds of thinned samples (one of twelve seeds examined gave $z = 3.5$ on its own; the standard deviation of z over seeds was 0.99, i.e. no bias), and a verification gate must not depend on the luck of one seed.

4.2. Identifiability

Structural. Table 3 gives the rank of the exact information matrix in the four observation configurations. With ratings only, six of the ten parameters that enter the likelihood are identifiable; four directions are null and the practical remedy is to fix A_0, α, σ_A and k_{int}, i.e. to give up the scale of both latent variables and the baseline. Adding one EEG channel leaves exactly one null direction, and in both cases it is an exact symmetry of the model (Appendix B): when Φ is not observed, $(k_{\mathrm{int}}, \sigma_\Phi, \alpha) \mapsto (c\,k_{\mathrm{int}}, c\,\sigma_\Phi, \alpha/c)$ leaves the law of all observations unchanged (the scale of Φ is arbitrary); when A is not observed, the affine map $A \mapsto cA + (1-c)A_{\max}$ with $(k_{\mathrm{int}}, A_0, \sigma_A, \beta, \gamma_0) \mapsto (k_{\mathrm{int}}/c,\ cA_0 + (1-c)A_{\max},\ c\sigma_A,\ \beta/c,\ \gamma_0 + \beta(1-c)/c)$ does the same (the scale of A is anchored only by $A_{\max}$). The numerically computed null vectors coincide with the gradients of these one-parameter families. Only configuration (d), ratings plus both EEG channels, identifies all 12 parameters, with a well-conditioned information matrix (condition number 188 after normalisation).

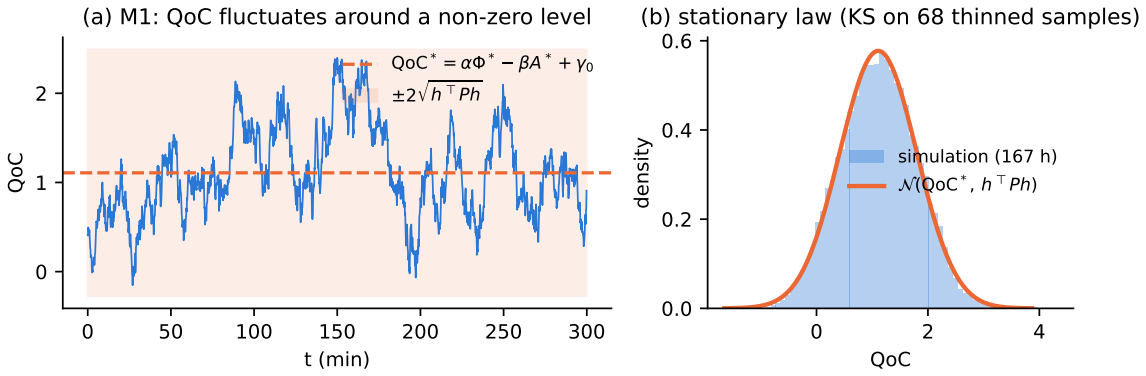


Figure 2: Propositions 2.4–2.5 for M1 at the nominal parameters. (a) A 300-min excerpt of a simulated session (exact discretisation, $\Delta = 10$ s) fluctuating around $\mathrm{QoC}^* = 1.108$; the band is $\pm 2\sqrt{\mathbf{h}^\top P \mathbf{h}}$. (b) Histogram of QoC over a 10^4-min simulation against the closed-form $\mathcal{N}(\mathrm{QoC}^*, \mathbf{h}^\top P \mathbf{h})$ (the panel title counts the thinned samples of this single run). The verification gate uses a Kolmogorov–Smirnov test on 2,000 thinned samples (one every five slowest time constants) pooled over five independent chains: $D = 0.020$, $p = 0.42$; errors of the sample mean and standard deviation 0.005 and 0.007.

Table 3: Structural identifiability by Rothenberg's rank criterion (90-min session, probes every minute, $\gamma_0 = 0.3$). "Absent": the parameter does not enter the likelihood. The null directions of (b) and (c) are the exact symmetries of Appendix B. The last column names the parameters that must be fixed to restore local identifiability.

Configuration	Absent	Rank	Null direction(s)	Fix
(a) ratings	r_A, r_Φ	6/10	four (symmetries below + noise scales)	$A_0, \alpha, \sigma_A, k_{\mathrm{int}}$
(b) ratings + y_A	r_Φ	10/11	Φ-scaling: $k_{\mathrm{int}}, \sigma_\Phi \uparrow, \alpha \downarrow$	α
(c) ratings + y_Φ	r_A	10/11	A-affine: $\gamma_0, \beta, A_0, k_{\mathrm{int}} \uparrow, \sigma_A \downarrow$	γ_0
(d) ratings + $y_A + y_\Phi$	–	12/12	none	none

Practical. Table 4 lists the relative CRB at the reference design. Two patterns matter for design. First, the three rate constants $k_{\mathrm{inh}}, k_{\mathrm{int}}, k_d$ are the hardest parameters in every configuration (15–18 % in configuration (d), against 3 % for A_0 and 4–7 % for the gains). Second, the EEG channels do most of the work: ratings alone give $\mathrm{CRB}(k_{\mathrm{inh}}) = 556\,\%$ and $\mathrm{CRB}(\beta) = 277\,\%$, i.e. no usable information, and a single channel already brings k_{inh} to 16–48 % depending on which channel it is. The DMN channel is the one that matters for k_{inh} and A_0 (configuration (b) is almost as good as (d) for them), the Φ channel for σ_Φ and α.

The profile likelihoods in Figure 3 confirm that the rate constants are practically identifiable in configuration (d): no profile is flat or multimodal in the ± 0.8 log-unit window, and the likelihood-based 95 % intervals have widths consistent with the Fisher bounds.

4.3. The bound predicts the error of maximum likelihood

Figure 4 compares the realised RMSE of the ML estimator with the CRB in 24 cells covering N from 6 to 48, sessions from 20 to 90 minutes, all probe intervals and all

Table 4: Relative Cramér–Rao bound by observation configuration at the reference design ($N = 12$, 45 min, probe every 2 min, $r_A = r_\Phi = 0.1$, $r_Q = 0.3$). "fixed": normalisation of a non-identifiable direction (Table 3); "absent": parameter does not enter the likelihood; for $\gamma_0 = 0$ the absolute standard error is given.

	(a) ratings	(b) ratings + y_A	(c) ratings + y_Φ	(d) ratings + $y_A + y_\Phi$
k_{inh}	5.559	0.157	0.480	0.155
A_0	0.325	0.031	0.170	0.030
σ_A	fixed	0.049	0.309	0.049
k_{int}	fixed	0.260	0.254	0.175
k_d	1.155	0.337	0.372	0.178
σ_Φ	8.607	0.175	0.063	0.057
α	fixed	fixed	0.092	0.037
β	2.765	0.116	0.137	0.072
γ_0	fixed	SE = 0.736	fixed	SE = 0.130
r_A	absent	0.016	absent	0.016
r_Φ	absent	absent	0.014	0.014
r_Q	0.212	0.070	0.122	0.050

Table 5: Cheapest design in the grid (smallest $N \times T$, ties broken by the bound) with relative CRB $\leq 20\,\%$ for each key parameter, and the fraction of the 144 grid cells that are adequate for it.

Parameter	N	session (min)	probe (min)	EEG noise r	CRB	adequate cells
k_{inh}	24	20	0.5	0.05	0.142	75 %
A_0	6	20	0.5	0.05	0.057	100 %
α	6	20	0.5	0.05	0.037	100 %
β	6	20	0.5	0.05	0.072	98 %

noise levels. The ratio RMSE/CRB has median 0.98 and per-parameter medians 0.96 (k_{inh}), 0.97 (A_0), 1.00 (α) and 1.06 (β); no ratio exceeds 2 and the five ratios below 0.5 (minimum 0.40) are the sampling noise of an RMSE computed from five fits falling *below* the bound, not evidence of a problem. The concern that the bound is optimistic at small N does not materialise: in the four $N = 6$ cells the ratios lie between 0.7 and 1.5. At the reference cell the bound predicts $(15.5, 3.0, 3.7, 7.2)$ % for $(k_{\text{inh}}, A_0, \alpha, \beta)$ and ML achieves $(19.7, 2.6, 3.8, 7.7)$ %. We therefore use the exact CRB, which costs about two seconds per design, in place of simulation for the full design map.

4.4. Design map

Figure 5 and Table 5 are the central practical result. Three facts stand out.

The DMN inhibition rate is the bottleneck. A_0 and α are adequately estimated in every one of the 144 designs, β in all but the three smallest (6 subjects, 20 min, probes every 4 min), whereas k_{inh} is adequate in 75 % of them and fails in every 6-subject design shorter than 90 minutes (worst case 38 % at $N = 6$, 20 min, $r = 0.2$). With probes every 2 minutes and 45-minute sessions, $N = 12$ reaches 14.3–15.5 % for k_{inh}

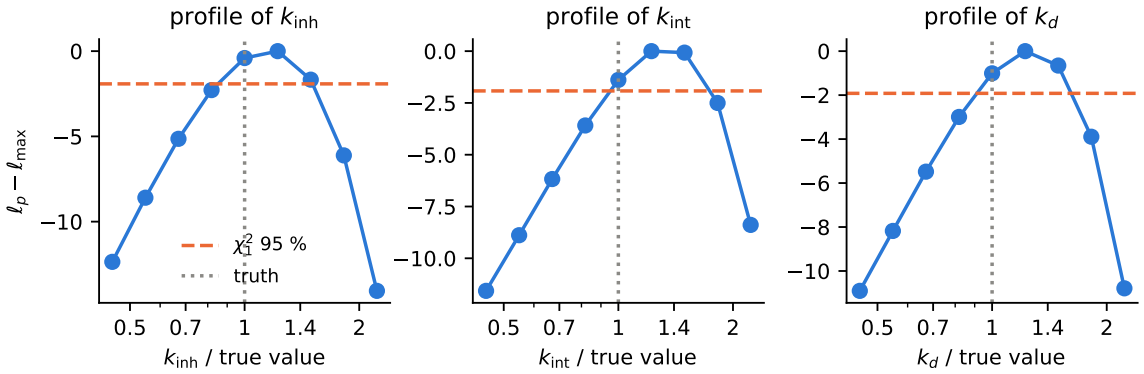


Figure 3: Profile log-likelihoods of the three rate constants on one simulated data set of the reference design (configuration (d)), all other parameters re-optimised at each grid point. Dashed line: χ^2_1 95 % threshold (-1.92); dotted line: true value. Each profile is unimodal with its maximum within one grid step of the truth (k_{inh}: 0.244 vs 0.200; k_{int}: 0.099 vs 0.081; k_d: 0.042 vs 0.034), and the curvature implied by the drop of 8–14 log-likelihood units at ± 0.8 log units corresponds to relative standard errors of 0.15–0.20, in line with the bounds of 0.16–0.18 in Table 4.

Table 6: Exponents of the power-law fit CRB $\propto N^a\, T^b\, r^c\, T_p^d$ over the 144 grid cells (log-linear least squares; N subjects, T session length, r the EEG noise sd, T_p the probe interval).

Parameter	a (subjects)	b (session length)	c (EEG noise)	d (probe interval)
k_{inh}	-0.50	-0.46	0.17	0.01
A_0	-0.50	-0.39	0.02	0.00
α	-0.50	-0.50	0.17	0.44
β	-0.50	-0.50	0.20	0.45

at $r \le 0.1$ and 18.3 % at $r = 0.2$, while $N = 6$ gives 22 %.

Simple scaling rules. A log-linear regression of the bound on the design variables over the 144 cells (Table 6) shows that every bound scales exactly as $N^{-1/2}$, as it must for independent subjects, and almost as $T^{-1/2}$ in the session length; the exponent of T is slightly weaker for k_{inh} (-0.46) and A_0 (-0.39) because each new subject contributes an independent draw from the stationary law whereas extra minutes of the same subject are autocorrelated. Many short sessions are therefore marginally better than few long ones for the DMN parameters and equivalent for the gains: at equal NT, 24 $\times$ 20 min gives CRB$(A_0) = 2.9$ % against 3.2 % for 6 $\times$ 90 min. The remaining design variables act differently on different parameters: the probe interval has no effect on k_{inh} and A_0 (which are informed by the dense EEG channel) but enters the gains α, β with exponent ≈ 0.45, i.e. halving the probe interval reduces their bound by about 27 %; the EEG noise affects k_{inh} and the gains with exponent ≈ 0.17–0.20 and A_0 not at all. For k_{inh} the rule reduces, to a good approximation, to a total-recording-time requirement: at $r \le 0.1$ about 480–540 subject-minutes of two-channel EEG (24 $\times$ 20 min, 12 $\times$ 45 min or 6 $\times$ 90 min) bring the bound to 15–16 %.

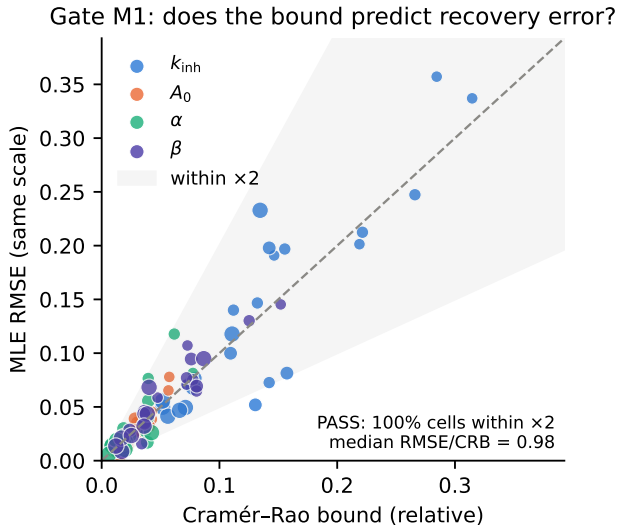


Figure 4: Gate M1. Root-mean-square error of maximum-likelihood recovery (five independent data sets per cell, 24 design cells spanning the grid) against the relative Cramér–Rao bound, both in the same scale; marker size grows with N. The shaded region is a factor of two around the identity. All 24 cells pass; the median ratio RMSE/CRB is 0.98 (10th–90th percentiles 0.61–1.40, extremes 0.40 and 1.93).

Table 7: Policy comparison on the true model (270 epochs, 20 trials, common random numbers). Cost $= \overline{(\mathrm{QoC} - \mathrm{QoC}_{\mathrm{tgt}})^2} + 0.1\,\overline{u^2}$. The last row is the closed-form stationary value of the unclipped LQG loop (Appendix F); the corresponding long simulation gives 0.0346 (difference 3 %).

Policy	Cost	Tracking MSE	RMS input
LQG (true parameters)	0.039	0.035	0.22
Bang-bang (same estimator)	0.561	0.162	2.00
No control	1.481	1.481	0
LQG, stationary closed form	0.034	0.030	0.18

Probe schedules. With 22 probes in a 45-minute session, the uniform schedule attained the smallest Ds-criterion among all schedules tested. The nearest competitors, mild front- or back-loading ($p = 0.85$ and 1.15), were within 0.05 % of it in geometric-mean standard error; strong front-loading ($p = 0.4$) cost 1.2 %, and paired probes one epoch apart cost 4.6 % (pairs 12 epochs apart 0.4 %). Strong back-loading ($p \geq 2.2$) collapses probes onto the same epochs and is infeasible. We make no claim of global optimality over all schedules; the result is that the common uniform schedule is not beaten by the usual non-uniform alternatives, because in configuration (d) the fast dynamics are already resolved by the EEG channels and the ratings mainly inform the static gains, for which spreading the probes evenly is natural.

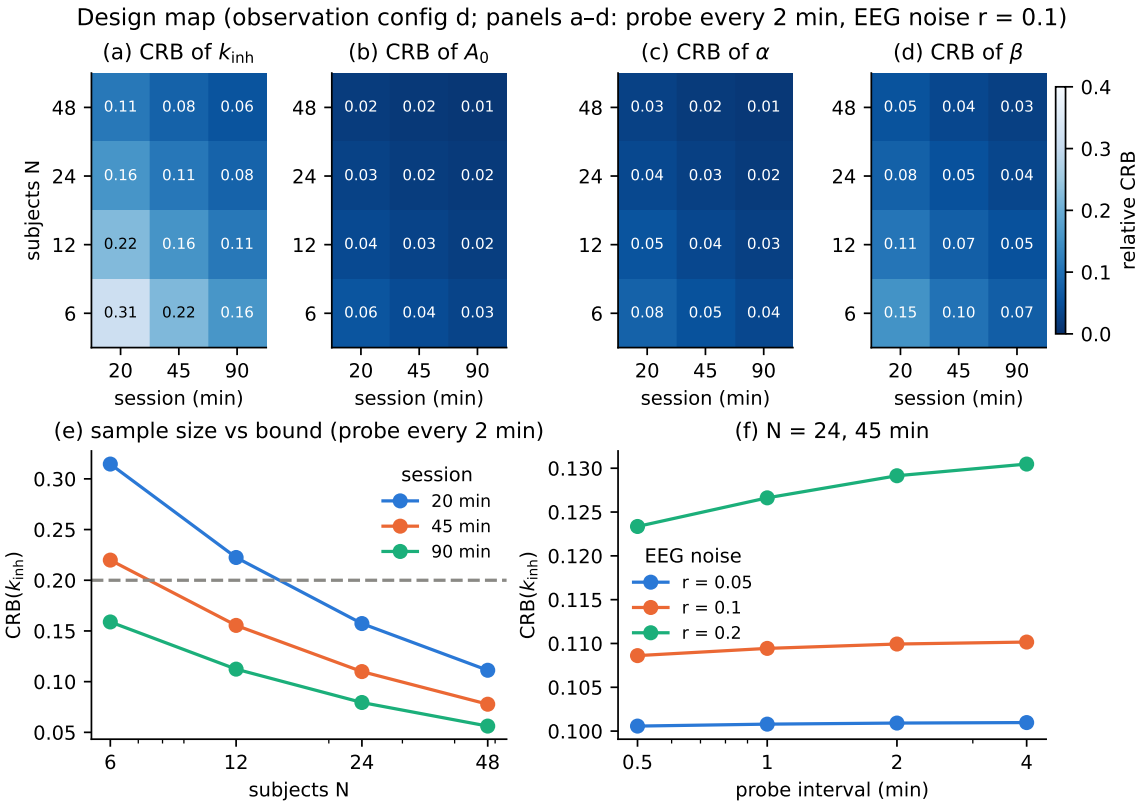


Figure 5: Design map (configuration (d)). (a–d) Relative CRB of k_{inh}, A_0, α, β as a function of the number of subjects and the session length, for probes every 2 min and EEG noise $r = 0.1$; cells with values below 0.2 are adequate. (e) CRB(k_{inh}) versus N for the three session lengths; the dashed line is the 20 % criterion. (f) Effect of the probe interval and of the EEG noise level on CRB(k_{inh}) at $N = 24$, 45 min: the probe interval is irrelevant for k_{inh}, the EEG noise is not.

4.5. Closed-loop control

Figure 6 and Table 7 summarise gate M2. With the true parameters, LQG achieves a session cost 14 times lower than bang-bang and 38 times lower than no control. The two neutral metrics tell the same story: LQG tracks the target with a mean-squared error of 0.035 (about 7 % of the open-loop variance) while using an input of RMS 0.22, whereas bang-bang tracks with an error of 0.16 at the price of saturating the actuator all the time. The required mean input is small ($u_{\text{ss}} = 0.024$) because the target is only 0.9 units above the open-loop level; the cost of LQG is dominated by the irreducible effect of process noise within one epoch, which is why its closed-form stationary cost (0.034) is close to the finite-horizon value. The comparison is made on the quadratic cost that LQG minimises; the point is not that bang-bang is a poor solution of *its* problem, but that a model-based regulator exploiting the estimated state and the known dynamics is far more efficient at holding a level, which is the natural objective of a neurofeedback loop.

Figure 7 connects the design map to control. When the controller is designed on parameters estimated from a design that is adequate for k_{inh} (19 of the 24 cells), the median loss of efficiency is a few per cent: 2.6 % (90th percentile 11 %) at the



Figure 6: One 45-minute closed-loop session on the true model with common random numbers for the three policies (target $\mathrm{QoC}_{\mathrm{tgt}} = 2.0$, $|u| \leq 2$). Top: QoC; bottom: input u. LQG holds QoC at the target with a small, smooth input; bang-bang chatters between $\pm u_{\max}$ and overshoots; without control QoC fluctuates around $\mathrm{QoC}^* = 1.1$.

reference design, 1.8 % (3.4 %) for 24 subjects × 45 min, below 1 % (90th percentile ≤ 2.1 %) for every 48-subject design at $r = 0.05$, and at most 7 % (90th percentile at most 17 %) over all adequate cells. The large losses all come from the five inadequate cells: 6 subjects × 20 min with probes every 2 min loses 38 % in the median and 55 % at the 90th percentile, and 12 subjects × 20 min at $r = 0.2$ has a median loss of 8 % but a 90th percentile of 98 %—one in ten controllers designed from such data would double the cost (two further inadequate cells have moderate medians but 90th percentiles of 28–32 %). A design that cannot estimate the model therefore cannot support a model-based neurofeedback loop built on it, and the 20 % criterion of Section 4.4 is, to a good approximation, the boundary between controllers that lose a few per cent and controllers that may lose most of their benefit.

4.6. Global sensitivity

Figure 8 and Table 8 show which parameters the conclusions depend on.

Identifiability of k_{inh} is set by the DMN baseline and its variability, not by k_{inh}. The total index of A_0 on $\log \mathrm{CRB}(k_{\mathrm{inh}})$ is 0.58 and that of CV_A is 0.36, while k_{inh} itself contributes 0.13 and the remaining five parameters nothing. The first-order indices (0.46 and 0.25) are smaller than the total ones, reflecting the interaction $A_0 \times \mathrm{CV}_A$: what the EEG channel resolves is the amplitude of the DMN fluctuations, $\mathrm{CV}_A \cdot A_0$, relative to the observation noise. For an experimenter this means that the sample-size requirement for k_{inh} depends on the signal-to-noise ratio of the DMN feature far

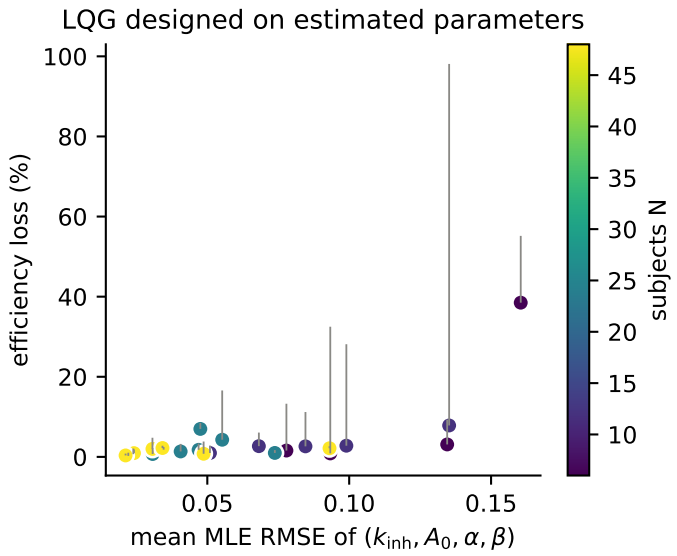


Figure 7: Robustness of LQG to parameter estimation error. For each of the 24 design cells, a controller (gain and filter) is designed on each of the five ML parameter sets recovered in that cell and run on the true model of that cell. Points: median efficiency loss over fits; whiskers extend to the 90th percentile; colour: number of subjects; horizontal axis: mean relative RMSE of $(k_{\text{inh}}, A_0, \alpha, \beta)$ in that cell. Full values in Table 9.

more than on the (unknown) value of the rate itself, and that improving the DMN feature is worth more than adding probes.

Stationary variability is governed by the gain α and the Φ noise. $\log \mathrm{Var}_\infty(\mathrm{QoC})$ has $S_T = 0.36$ for α, 0.26 for σ_Φ and 0.19 for A_0; the indices are close to additive ($S_1 \approx S_T$) and converged ($\max|\Delta S_T| = 0.013$).

Control efficiency has no dominant parameter. The LQG cost ratio spreads its sensitivity over $\beta, \alpha, A_0, k_{\text{inh}}, k_d$ (S_T from 0.17 to 0.29) with small first-order indices, i.e. the benefit of closed-loop control is a joint property of the gains and the DMN dynamics; no single parameter can make or break it within the range explored.

The stationary level QoC^* depends on $A_0, k_d, k_{\text{int}}, \alpha$ and not on β (because $\mathrm{QoC}^* = \alpha k_{\text{int}}(1 - A_0)/k_d - \beta A_0$ and the first term dominates). Its indices sum to more than one and have wide intervals (convergence 0.10) because the output is a product of inputs on the raw scale; we only retain the ordering.

5. Discussion

A consistency check that should precede fitting. The first result is negative but, we think, instructive beyond the particular model: a differential equation in which the "conscious" variable is driven only by the time derivatives of its inputs cannot represent a sustained state, whatever its parameters. Substituting a proposed steady state back into the model equations (Proposition 2.3) is a short calculation that we recommend as a routine step before any such model is fitted. The repaired

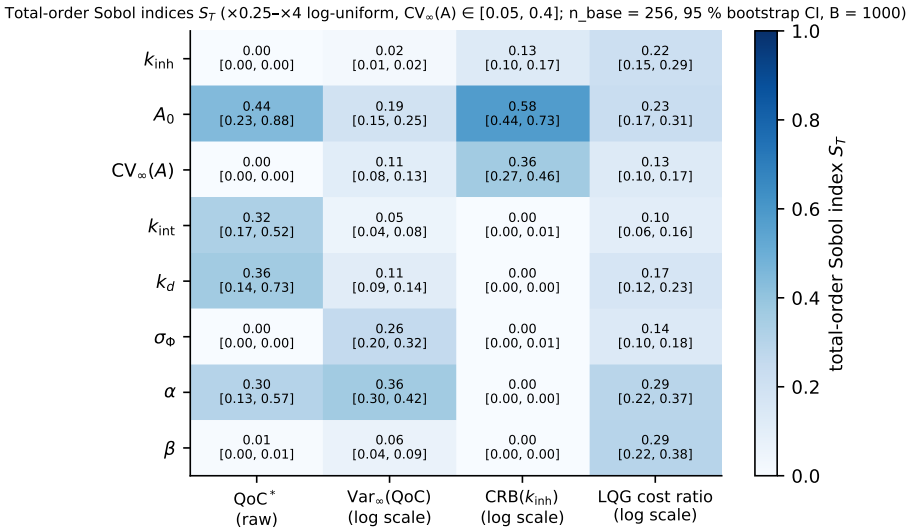


Figure 8: Total-order Sobol indices S_T (with 95 % bootstrap intervals) of four outputs with respect to eight model parameters varied log-uniformly over $\times 0.25$–$\times 4$ around the nominal point ($\mathrm{CV}_A \in [0.05, 0.4]$). $\mathrm{Var}_\infty(\mathrm{QoC})$, $\mathrm{CRB}(k_{\mathrm{inh}})$ and the LQG efficiency are analysed on the log scale; QoC^* on the raw scale. Scrambled Sobol sampling, $n = 256$, 2,560 evaluations.

Table 8: Total-order Sobol indices S_T with 95 % bootstrap confidence intervals ($B = 1{,}000$). The last row is the convergence diagnostic $\max_i |\Delta S_{T,i}|$ between the full sample and its first half.

Input	QoC^* (raw)	$\log \mathrm{Var}_\infty(\mathrm{QoC})$	$\log \mathrm{CRB}(k_{\mathrm{inh}})$	log LQG efficiency
k_{inh}	0.00	0.02 [0.01, 0.02]	0.13 [0.10, 0.17]	0.22 [0.15, 0.29]
A_0	0.44 [0.23, 0.88]	0.19 [0.15, 0.25]	**0.58** [0.44, 0.73]	0.23 [0.17, 0.31]
CV_A	0.00	0.11 [0.08, 0.13]	**0.36** [0.27, 0.46]	0.13 [0.10, 0.17]
k_{int}	0.32 [0.17, 0.52]	0.05 [0.04, 0.08]	0.00	0.10 [0.06, 0.16]
k_d	0.36 [0.14, 0.73]	0.11 [0.09, 0.14]	0.00	0.17 [0.12, 0.23]
σ_Φ	0.00	0.26 [0.20, 0.32]	0.00	0.14 [0.10, 0.18]
α	0.30 [0.13, 0.57]	**0.36** [0.30, 0.42]	0.00	0.29 [0.22, 0.37]
β	0.01 [0.00, 0.01]	0.06 [0.04, 0.09]	0.00	0.29 [0.22, 0.38]
$\max \lvert\Delta S_T\rvert$ (half sample)	0.10	0.013	0.076	0.030

model M1 keeps the mechanistic story of M0—practice suppresses the DMN, DMN disengagement feeds integration, both feed the reported quality—but turns it into a well-posed stationary stochastic process with closed-form moments. That it is linear-Gaussian is a modelling choice with a large pay-off: it makes the likelihood, the Fisher information and the optimal controller exact, which is what allows a complete design analysis without simulation.

What experience-sampling designs can and cannot deliver. The identifiability analysis gives a clear hierarchy. Ratings alone are not enough to learn the dynamics: without an EEG channel the scales of both latent variables are arbitrary and the rate constants are known to within factors of several. One EEG channel removes all but one exact symmetry, and the symmetry is interpretable (the unobserved variable can

be rescaled). Both channels identify everything. Within the identifiable configuration, the design map and the scaling rules of Table 6 translate into advice an experimenter can use directly: (i) the number of probes governs the precision of the gains that link the latent state to the report, with a square-root-like return; (ii) the precision of the DMN rate constant depends on the total amount of EEG recorded and on the quality of the DMN feature, and not on how often one probes; (iii) about eight subject-hours of two-channel EEG at a feature noise of 10 % of the signal scale bring the rate constant to a 15 % relative error, and six-subject studies with sessions under 90 minutes cannot; (iv) uniform probe schedules are a safe default. These statements are about a model, not about brains, but they are exactly the kind of statement that must be available before a study is run if the study is meant to estimate a dynamical model, and they are absent from the current literature on computational models of meditation [21, 33], which has so far been descriptive.

From estimation to intervention. The control analysis shows what a model buys once it is identified. A regulator that uses the state estimate and the known dynamics holds the reported quantity at a target with an input an order of magnitude smaller than a threshold policy and a tracking error more than four times smaller, and its performance can be predicted in closed form. More importantly, Figure 7 quantifies the cost of estimation error in the currency of the application: controllers designed from adequate designs lose a few per cent, controllers designed from inadequate designs may lose most of their benefit. This gives the 20 % identifiability criterion an operational meaning and argues for treating the design of the estimation study and the design of the neurofeedback protocol as one problem, in line with recent calls for more rigorous neurofeedback methodology [30, 34].

Robustness of the conclusions. The sensitivity analysis shows that the difficulty of estimating k_{inh} is not a peculiarity of the nominal point: across a 16-fold range of every parameter it is controlled by the DMN baseline and coefficient of variation, both of which are properties of the EEG feature rather than of the rate. The conclusions about control efficiency and stationary variability are likewise spread over several parameters rather than hinging on one.

5.1. Limitations

- **No real data are fitted.** The paper answers the question that precedes data analysis—which designs suffice to fit the model—and makes no claim about parameter values in practitioners. The nominal point is a working point; the sensitivity analysis covers a wide range around it but not qualitatively different regimes.
- **Linearity and Gaussianity.** M1 is linear with additive Gaussian noise and a linear read-out, and ratings are treated as continuous. Real ratings are ordinal

and bounded, the DMN level is confined to $[0, A_{\max}]$, and meditation dynamics may involve discrete state switches [9, 17]. Non-linear extensions (saturating read-out, state-dependent noise, regime switching) would require approximate filters and simulation-based Fisher information; the present analysis is the exact baseline against which such extensions can be compared.

- **The mapping of EEG features to** (A, Φ) is assumed, with noise levels as design variables. Establishing such a mapping is an empirical task; our results say how good it must be ($r \lesssim 0.1$–0.2 of the signal scale) for the dynamics to be identifiable.
- **The design map is expressed in minutes at the nominal rates.** Session lengths of 20–90 min correspond to 4–18 DMN time constants at $k_{\text{inh}} = 0.2$ but to only 0.9–3.9 at the printed value 0.043 min^{-1}. The sensitivity analysis shows that $\text{CRB}(k_{\text{inh}})$ depends only weakly on k_{inh} itself within $\times 0.25$–$\times 4$ ($S_T = 0.13$), but a reader who expects much slower DMN dynamics should rescale session length in units of $1/k_{\text{inh}}$ and expect the requirement to shift towards longer sessions.
- **Probe burden is not costed.** Dense probing (every 30 s) improves the gains in the map but interrupts the practice it is meant to measure; the map should be read together with a ceiling on the number of probes that a protocol tolerates, which is outside the model.
- **Probe-schedule optimisation** was restricted to two interpretable families; a free optimisation over probe positions could find better schedules, although the small gaps we observe suggest limited room for improvement in configuration (d).
- **Control** assumes an additive input to the DMN equation with a known gain, perfect actuation within bounds and no delay, and compares policies on a quadratic cost. The robustness analysis covers parameter error but not model-structure error.
- **Between-subject heterogeneity** is not modelled: subjects share one parameter vector. A hierarchical extension would change the design map quantitatively (more subjects, shorter sessions) but not the identifiability structure.

6. Conclusion

For a recently proposed DMN–Φ–QoC model of meditation dynamics we have shown that the original formulation has no sustained QoC, proposed a linear-Gaussian replacement with closed-form stationary behaviour, established exactly which observation configurations identify its parameters, validated the Cramér–Rao bound against maximum likelihood across the design space, derived a quantitative design map and scaling rules for the number of subjects, session length, probe inter-

val and EEG quality, formulated and solved the closed-loop control problem, and quantified how estimation error from each design propagates to control performance. The combination of exact Fisher information, machine-checked verification gates and a single reproducible command is, we believe, a template for how dynamical models of mental states should be examined before they are taken to data.

Data and code availability

No empirical data were used. All results are generated *in silico* by the accompanying Python code (modules `bqi/dmn_ode.py`, `model/`, `eeg/model_fit.py` and `experiments/run_model_paper.py`) with seed 42; the full configuration snapshot, raw results, gate records and figures are included in the repository under `outputs/`. The repository and a frozen archive with a DOI will be made public upon acceptance.

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Author contributions

D.V. conceived the study, developed the methodology and software, performed the analysis and wrote the manuscript.

Declaration of competing interest

The author declares no competing interests. The author is also the author of the original model examined in this paper [40]; the corrections in Section 2.2 are reported here as part of the present analysis.

A. Proofs of Propositions 2.1–2.5

Proof of Proposition 2.1. At an equilibrium with constant drive $\bar{\xi}$, (1) gives $A^* = k_{\text{spont}}\bar{\xi}/k_{\text{inh}}$ and (2) gives $\Phi^* = (k_{\text{int}}/k_d)(1 - A^*/A_{\max})$. With $\dot{A} = \dot{\Phi} = 0$, (3) reduces to $0 = -\gamma \text{QoC}^*$, hence $\text{QoC}^* = 0$ since $\gamma > 0$. The linear system is lower-triangular with negative diagonal $(-k_{\text{inh}}, -k_d, -\gamma)$, so the equilibrium is unique and asymptotically stable. □

Proof of Proposition 2.2. Write $z = \text{QoC} - v$. From (3), $\dot{z} = \dot{\text{QoC}} - \dot{v} = -\gamma \text{QoC} = -\gamma(z + v)$, a linear equation whose solution is $z(t) = e^{-\gamma t} z(0) - \gamma \int_0^t e^{-\gamma(t-s)} v(s)\, \mathrm{d}s$; adding $v(t)$ gives (6). Taking Laplace transforms of $\dot{\text{QoC}} = \dot{v} - \gamma \text{QoC}$ with zero initial conditions, $s\,\widehat{\text{QoC}} = s\hat{v} - \gamma \widehat{\text{QoC}}$, hence $H(s) = s/(s + \gamma)$, $H(0) = 0$ and

$|H(j\omega)|^2 = \omega^2/(\omega^2+\gamma^2)$. If $v(t) \to v_\infty$, the convolution term in (6) tends to v_∞ (since $\gamma \int_0^t e^{-\gamma(t-s)}\,\mathrm{d}s \to 1$), so $\mathrm{QoC}(t) \to v_\infty - v_\infty = 0$. □

Proof of Proposition 2.3. Substituting any triple with $\dot{A} = \dot{\Phi} = 0$ into (3) yields $\dot{\mathrm{QoC}} = -\gamma\mathrm{QoC}$, which vanishes only at $\mathrm{QoC} = 0$; by (4), $\mathrm{QoC}^*_{\mathrm{p}} > 0$ whenever $\alpha, \beta, k_{\mathrm{int}} > 0$ and $A^*_{\mathrm{p}} < A_{\max}$. The numerical residual follows by substituting the printed parameters with $\bar{\xi} = 0$: $\dot{A} = -k_{\mathrm{inh}} A^*_{\mathrm{p}} = -0.043 \times 0.0787 = -3.4 \times 10^{-3}$, $\dot{\Phi} = 0$ by construction of Φ^*_{p}, and $\dot{\mathrm{QoC}} = -\beta\dot{A} - \gamma\mathrm{QoC}^*_{\mathrm{p}} = 0.0063 - 0.19 \times 27.81 = -5.277$. □

Proof of Proposition 2.4. For the linear SDE $\mathrm{d}\mathbf{x} = (M\mathbf{x} + \mathbf{u})\,\mathrm{d}t + G\,\mathrm{d}\mathbf{W}$ with Hurwitz M, the mean obeys $\dot{\boldsymbol{\mu}} = M\boldsymbol{\mu} + \mathbf{u}$ and the covariance $\dot{P} = MP + PM^\top + GG^\top$ [13]; the stationary values solve $M\boldsymbol{\mu} + \mathbf{u} = 0$ and $MP + PM^\top + GG^\top = 0$. Solving the first with (10) gives (11). Writing out the Lyapunov equation entry-wise with M lower-triangular: the $(1,1)$ entry gives $-2k_{\mathrm{inh}}P_{AA} + \sigma_A^2 = 0$; the $(1,2)$ entry gives $-k_{\mathrm{inh}}P_{A\Phi} - (k_{\mathrm{int}}/A_{\max})P_{AA} - k_d P_{A\Phi} = 0$; the $(2,2)$ entry gives $-2(k_{\mathrm{int}}/A_{\max})P_{A\Phi} - 2k_d P_{\Phi\Phi} + \sigma_\Phi^2 = 0$. Solving in this order yields (12). At the nominal parameters $P_{AA} = 0.025$, $P_{A\Phi} = -0.00865$, $P_{\Phi\Phi} = 0.0574$ and $\mathbf{h}^\top P\mathbf{h} = \beta^2 P_{AA} - 2\alpha\beta P_{A\Phi} + \alpha^2 P_{\Phi\Phi} = 0.477$. $\mathrm{QoC}^* = 0$ defines a codimension-one set in parameter space, so $\mathrm{QoC}^* \neq 0$ generically. □

Proof of Proposition 2.5. The solution of a linear SDE driven by Brownian motion from a Gaussian initial condition is a Gaussian process; with the stationary initial law it is stationary. For $\tau \geq 0$, $\mathbf{x}(t+\tau) = e^{M\tau}\mathbf{x}(t) + \int_t^{t+\tau} e^{M(t+\tau-s)}(\mathbf{u}\,\mathrm{d}s + G\,\mathrm{d}\mathbf{W}_s)$, and the integral is independent of $\mathbf{x}(t)$, so $\mathrm{Cov}(\mathbf{x}(t+\tau), \mathbf{x}(t)) = e^{M\tau}P$. A linear functional $\mathbf{h}^\top\mathbf{x} + \gamma_0$ of a Gaussian process is Gaussian with the stated mean, variance and autocovariance. □

B. Exact symmetries behind the null directions

Let $\mathcal{L}_{\boldsymbol{\theta}}$ denote the joint law of all observations of a session.

Proposition B.1 (Φ-scaling, configuration (b))**.** *If y_Φ is not observed, then for every $c > 0$ the parameter map $k_{\mathrm{int}} \mapsto ck_{\mathrm{int}}$, $\sigma_\Phi \mapsto c\sigma_\Phi$, $\alpha \mapsto \alpha/c$ (all other parameters fixed) leaves $\mathcal{L}_{\boldsymbol{\theta}}$ unchanged.*

Proof. Set $\Phi' = c\Phi$. Then $\mathrm{d}\Phi' = [ck_{\mathrm{int}}(1 - A/A_{\max}) - k_d\Phi']\,\mathrm{d}t + c\sigma_\Phi\,\mathrm{d}W_\Phi$, which is (8) with $(ck_{\mathrm{int}}, c\sigma_\Phi)$; the A equation is untouched; and $\mathrm{QoC} = \alpha\Phi - \beta A + \gamma_0 = (\alpha/c)\Phi' - \beta A + \gamma_0$. The observed processes A (through y_A) and QoC (through ratings) therefore have the same law under both parameter vectors. □

Proposition B.2 (A-affine map, configuration (c))**.** *If y_A is not observed, then for every*

$c > 0$ the map

$$k_{\text{int}} \mapsto k_{\text{int}}/c, \qquad A_0 \mapsto cA_0 + (1-c)A_{\max}, \qquad \sigma_A \mapsto c\sigma_A,$$
$$\beta \mapsto \beta/c, \qquad \gamma_0 \mapsto \gamma_0 + \beta(1-c)/c$$

leaves $\mathcal{L}_{\boldsymbol{\theta}}$ unchanged.

Proof. Set $A' = cA + (1-c)A_{\max}$. Then $\mathrm{d}A' = c\,\mathrm{d}A = -k_{\text{inh}}\big(A' - [cA_0 + (1-c)A_{\max}]\big)\,\mathrm{d}t + c\sigma_A\,\mathrm{d}W_A$. In the Φ equation, $k_{\text{int}}(1 - A/A_{\max}) = k_{\text{int}}\big(1 - (A' - (1-c)A_{\max})/(cA_{\max})\big) = (k_{\text{int}}/c)(1 - A'/A_{\max})$. In the read-out, $-\beta A = -(\beta/c)A' + \beta(1-c)A_{\max}/c$, absorbed into γ_0 (with $A_{\max} = 1$). The observed processes Φ and QoC have the same law under both parameter vectors. □

Differentiating each family at $c = 1$ in the transformed coordinates (log for positive parameters) gives the tangent directions $(\mathrm{d}\log k_{\text{int}},\, \mathrm{d}\log\sigma_\Phi,\, \mathrm{d}\alpha) = (1, 1, -\alpha)$ and $(\mathrm{d}\log k_{\text{int}},\, \mathrm{d}A_0,\, \mathrm{d}\log\sigma_A,\, \mathrm{d}\beta,\, \mathrm{d}\gamma_0) = (-1,\, A_0 - A_{\max},\, 1,\, -\beta,\, -\beta)$. For each tangent $\mathbf{v}$ the quadratic form $\mathbf{v}^\top\mathcal{I}\mathbf{v}$ vanishes to machine precision (10^{-16} relative to $\mathbf{v}^\top\,\mathrm{diag}(\mathcal{I})\mathbf{v}$), and after the diagonal normalisation of Section 3.4 the tangent coincides with the numerically found null vector (cosine 1.000): in configuration (b) the normalised null vector is $(\alpha, k_{\text{int}}, \sigma_\Phi) = (0.709, -0.619, -0.337)$, in configuration (c) $(\gamma_0, \beta, A_0, k_{\text{int}}, \sigma_A) = (0.767, 0.491, 0.281, 0.250, -0.170)$, up to sign. In configuration (a) both families are available, together with two further directions that trade the noise scales σ_A, σ_Φ against the gains; fixing $(A_0, \alpha, \sigma_A, k_{\text{int}})$ removes all four. In configuration (d) both A and Φ are observed directly, no rescaling is possible, and the information matrix has full rank.

C. Van Loan discretisation

For $n = 2$, form the $2n \times 2n$ matrix $V = \begin{pmatrix} -M & GG^\top \\ 0 & M^\top \end{pmatrix}$ and compute $E = e^{V\Delta} = \begin{pmatrix} E_{11} & E_{12} \\ 0 & E_{22} \end{pmatrix}$. Then $F = E_{22}^\top = e^{M\Delta}$ and $Q = E_{22}^\top E_{12} = \int_0^\Delta e^{Ms} GG^\top e^{M^\top s}\,\mathrm{d}s$ [37]. The deterministic input is integrated exactly as $\mathbf{c} = M^{-1}(F - I)\mathbf{u}$, and the control input matrix as $B = M^{-1}(F - I)\mathbf{b}$ (zero-order hold). The discrete-time stationary covariance satisfies $P = FPF^\top + Q$ and coincides with the continuous-time P of (12), which is used as a unit test.

D. Sequential Kalman filter

With prior $\hat{\mathbf{x}}_{0|0} = \boldsymbol{\mu}$, $P_{0|0} = P$, each epoch consists of the prediction $\hat{\mathbf{x}}_{t|t-1} = F\hat{\mathbf{x}}_{t-1|t-1} + \mathbf{c}$, $P_{t|t-1} = FP_{t-1|t-1}F^\top + Q$, followed by one scalar update per available observation $y = \mathbf{s}^\top\mathbf{x} + o + e$, $e \sim \mathcal{N}(0, r^2)$, with $(\mathbf{s}, o, r) = (\mathbf{e}_1, 0, r_A)$ for y_A, $(\mathbf{e}_2, 0, r_\Phi)$

for y_Φ and $(\mathbf{h}, \gamma_0, r_Q)$ for a rating:

$$\nu = y - \mathbf{s}^\top \hat{\mathbf{x}} - o, \qquad S = \mathbf{s}^\top P \mathbf{s} + r^2, \qquad \mathbf{k} = P\mathbf{s}/S,$$
$$\hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}} + \mathbf{k}\nu, \qquad P \leftarrow P - \mathbf{k}\mathbf{k}^\top S, \qquad \ell \mathrel{+}= -\tfrac{1}{2}\big(\log 2\pi S + \nu^2/S\big).$$

Because the observation noises are independent, sequential scalar updates are exactly equivalent to the vector update [11]; the total ℓ is the exact log-likelihood of the session and equals the Gaussian log-density of the stacked vector of Appendix E.

E. Exact Fisher information of a session

Let T be the number of epochs and stack the latent path $\mathbf{X} = (\mathbf{x}_1^\top, \dots, \mathbf{x}_T^\top)^\top \in \mathbb{R}^{2T}$. Under stationarity $\mathbb{E}\mathbf{X} = \mathbf{1}_T \otimes \boldsymbol{\mu}$ and $\mathrm{Cov}(\mathbf{x}_t, \mathbf{x}_s) = F^{t-s}P$ for $t \geq s$ (and its transpose for $t < s$), so Σ_X is block-Toeplitz with blocks $C_k = F^k P$. Every observation is $y = \mathbf{s}^\top \mathbf{X} + o + e$ for a selection row $\mathbf{s}$ (a unit vector for y_A, y_Φ, the read-out $\mathbf{h}$ placed at the probe epoch for a rating); collecting the rows into S, offsets into $\mathbf{o}$ and noise variances into D,

$$\mathbf{Y} = S\mathbf{X} + \mathbf{o} + \mathbf{e} \sim \mathcal{N}\big(\mathbf{m}, \Sigma\big), \qquad \mathbf{m} = S(\mathbf{1} \otimes \boldsymbol{\mu}) + \mathbf{o}, \qquad \Sigma = S\Sigma_X S^\top + D.$$

Equation (17) is the standard Fisher information of a Gaussian vector with parameter-dependent mean and covariance [22]. For the reference design Σ is 562×562 and (17) takes about two seconds for all 12 parameters. For N subjects with the same design the information is $N\mathcal{I}$; the relative CRB of parameter i is $\sqrt{[(N\mathcal{I})^{-1}]_{ii}}$ in the transformed coordinates (divided by $|\theta_i|$ for linear-scale parameters). Rothenberg's criterion is applied to $\mathcal{I}$ normalised by its diagonal, $\tilde{\mathcal{I}} = D^{-1/2}\mathcal{I}D^{-1/2}$; the relative eigenvalue threshold 10^{-7} separates eigenvalues of order 10^{-16} (exact symmetries) from the smallest genuine eigenvalue (5×10^{-3} in configuration (d)).

F. LQG details and closed-form stationary cost

With $G = (I - F)^{-1}$ the equilibrium is $u_{\mathrm{ss}} = (\mathrm{QoC}_{\mathrm{tgt}} - \gamma_0 - \mathbf{h}^\top G\mathbf{c})/(\mathbf{h}^\top GB)$, $\mathbf{x}_{\mathrm{ss}} = G(\mathbf{c} + Bu_{\mathrm{ss}})$. The LQR gain is $K = (\lambda + B^\top \Pi B)^{-1} B^\top \Pi F$ with Π the stabilising solution of the discrete algebraic Riccati equation for $(F, B, \mathbf{h}\mathbf{h}^\top, \lambda)$; the stationary Kalman gain is $L = P_-(P_- + R)^{-1}$ with P_- the solution of the dual Riccati equation for $(F^\top, I, Q, R)$, $R = \mathrm{diag}(r_A^2, r_\Phi^2)$. Without clipping, the loop $u_t = u_{\mathrm{ss}} - K(\hat{\mathbf{x}}_t - \mathbf{x}_{\mathrm{ss}})$, $\hat{\mathbf{x}}_{t+1} = (I - L)(F\hat{\mathbf{x}}_t + \mathbf{c} + Bu_t) + L\mathbf{y}_{t+1}$ makes $\mathbf{z}_t = (\mathbf{x}_t, \hat{\mathbf{x}}_t)$ a linear Gaussian recursion

$\mathbf{z}_{t+1} = \mathcal{A}\mathbf{z}_t + \mathbf{d} + \boldsymbol{\zeta}_t$ with

$$\mathcal{A} = \begin{pmatrix} F & -BK \\ LF & (I-L)F - BK \end{pmatrix}, \qquad \mathrm{Cov}(\boldsymbol{\zeta}) = \begin{pmatrix} Q & QL^\top \\ LQ & LQL^\top + LRL^\top \end{pmatrix},$$
$$\mathbf{d} = \begin{pmatrix} \mathbf{d}_0 \\ \mathbf{d}_0 \end{pmatrix}, \qquad \mathbf{d}_0 = \mathbf{c} + Bu_{\mathrm{ss}} + BK\mathbf{x}_{\mathrm{ss}}.$$

Its stationary mean $\mathbf{m}_z = (I - \mathcal{A})^{-1}\mathbf{d}$ and covariance $\mathcal{S}$ (discrete Lyapunov equation $\mathcal{S} = \mathcal{A}\mathcal{S}\mathcal{A}^\top + \mathrm{Cov}(\boldsymbol{\zeta})$) give the stationary cost

$$J_\infty = \left(\mathbf{h}^\top \mathbf{m}_x + \gamma_0 - \mathrm{QoC}_{\mathrm{tgt}}\right)^2 + \mathbf{h}^\top \mathcal{S}_{xx}\mathbf{h} + \lambda\left[\bar{u}^2 + K\mathcal{S}_{\hat{x}\hat{x}}K^\top\right],$$
$$\bar{u} = u_{\mathrm{ss}} - K(\mathbf{m}_{\hat{x}} - \mathbf{x}_{\mathrm{ss}}).$$

At the nominal parameters $J_\infty = 0.0336$ (tracking 0.0302, RMS input 0.185), against 0.0346 from a 4,000-epoch simulation with 1,000 epochs of burn-in; the loop is stable ($\rho(\mathcal{A}) < 1$). In the robustness analysis the controller's $(F, \mathbf{c}, B, \mathbf{h}, \gamma_0, K, L, \mathbf{x}_{\mathrm{ss}}, u_{\mathrm{ss}})$ are all computed from the estimated parameters, while the plant uses the true ones.

G. Robustness table

H. Reproduction

```
pip install -r requirements.txt
set OMP_NUM_THREADS=1   (Linux/macOS: export OMP_NUM_THREADS=1)
set MKL_NUM_THREADS=1
python main.py model all --seed 42 --n-base 256 --profiles   # ≈ 8 h on 4
cores
python main.py gates   # M0 PASS, M1 PASS, M2 PASS
python main.py model figures --profiles   # outputs/paper/model/fig1...fig8,
table1...table3
```

Individual packages: `model theory | identifiability | design | control | sensitivity`. The design package (24 cells $\times$ 5 maximum-likelihood fits) accounts for almost all of the running time; every other package runs in minutes. Results are bit-identical across machines with the same seed. The thread-count variables prevent oversubscription when the design and sensitivity packages run their worker processes.

References

[1] Anderson, B.D.O. and Moore, J.B. (1990). *Optimal Control: Linear Quadratic Methods*. Prentice Hall, Englewood Cliffs.

[2] Åström, K.J. (1970). *Introduction to Stochastic Control Theory*. Academic Press, New York.

Table 9: Efficiency loss of LQG designed on estimated parameters, by design cell (configuration (d), target 2.0). "RMSE": mean relative ML error of $(k_{\text{inh}}, A_0, \alpha, \beta)$ over the five fits of the cell; "cost (true)": LQG cost with true parameters at that cell's EEG noise level; loss in per cent, median and 90th percentile over the five fits.

N	session (min)	probe (min)	r	RMSE	cost (true)	loss median (%)	loss q90 (%)
6	20	0.5	0.05	0.135	0.0242	3.1	9.3
6	20	2.0	0.10	0.160	0.0350	38.5	55.2
6	45	1.0	0.10	0.093	0.0350	0.9	32.5
6	90	1.0	0.05	0.078	0.0242	1.6	13.3
12	20	1.0	0.10	0.099	0.0350	2.8	28.1
12	20	2.0	0.20	0.135	0.0560	7.8	98.1
12	45	2.0	0.10	0.085	0.0350	2.6	11.2
12	45	0.5	0.05	0.035	0.0242	1.8	2.5
12	90	1.0	0.10	0.051	0.0350	1.0	2.8
12	90	4.0	0.20	0.068	0.0560	2.7	6.1
24	20	0.5	0.05	0.074	0.0242	1.0	1.7
24	20	2.0	0.10	0.055	0.0350	4.3	16.6
24	45	1.0	0.10	0.047	0.0350	1.8	3.4
24	45	4.0	0.20	0.048	0.0560	7.0	8.5
24	90	1.0	0.05	0.031	0.0242	0.7	2.2
24	90	2.0	0.10	0.041	0.0350	1.4	3.2
48	20	1.0	0.10	0.049	0.0350	0.8	3.8
48	20	4.0	0.20	0.093	0.0560	2.2	2.7
48	45	0.5	0.05	0.022	0.0242	0.2	1.0
48	45	2.0	0.10	0.031	0.0350	2.0	4.8
48	90	1.0	0.05	0.024	0.0242	0.7	2.1
48	90	2.0	0.10	0.024	0.0350	0.9	1.8
48	90	4.0	0.20	0.034	0.0560	2.2	2.8
48	90	0.5	0.05	0.021	0.0242	0.4	0.8

[3] Atkinson, A.C., Donev, A.N. and Tobias, R.D. (2007). *Optimum Experimental Designs, with SAS*. Oxford University Press, Oxford.

[4] Bellman, R. and Åström, K.J. (1970). On structural identifiability. *Mathematical Biosciences*, 7(3–4), 329–339.

[5] Brandmeyer, T. and Delorme, A. (2018). Reduced mind wandering in experienced meditators and associated EEG correlates. *Experimental Brain Research*, 236, 2519–2528.

[6] Brewer, J.A., Worhunsky, P.D., Gray, J.R., Tang, Y.-Y., Weber, J. and Kober, H. (2011). Meditation experience is associated with differences in default mode network activity and connectivity. *Proceedings of the National Academy of Sciences*, 108(50), 20254–20259.

[7] Byrd, R.H., Lu, P., Nocedal, J. and Zhu, C. (1995). A limited memory algorithm for bound constrained optimization. *SIAM Journal on Scientific Computing*, 16(5), 1190–1208.

[8] Chis, O.-T., Banga, J.R. and Balsa-Canto, E. (2011). Structural identifiability of systems biology models: a critical comparison of methods. *PLoS ONE*, 6(11), e27755.

[9] Christoff, K., Irving, Z.C., Fox, K.C.R., Spreng, R.N. and Andrews-Hanna, J.R. (2016). Mind-wandering as spontaneous thought: a dynamic framework. *Nature Reviews Neuroscience*, 17, 718–731.

[10] Cramér, H. (1946). *Mathematical Methods of Statistics*. Princeton University Press, Princeton.

[11] Durbin, J. and Koopman, S.J. (2012). *Time Series Analysis by State Space Methods* (2nd ed.). Oxford University Press, Oxford.

[12] Fox, K.C.R., Dixon, M.L., Nijeboer, S., Girn, M., Floman, J.L., Lifshitz, M., Ellamil, M., Sedlmeier, P. and Christoff, K. (2016). Functional neuroanatomy of meditation: a review and meta-analysis of 78 functional neuroimaging investigations. *Neuroscience & Biobehavioral Reviews*, 65, 208–228.

[13] Gardiner, C.W. (2009). *Stochastic Methods: A Handbook for the Natural and Social Sciences* (4th ed.). Springer, Berlin.

[14] Garrison, K.A., Zeffiro, T.A., Scheinost, D., Constable, R.T. and Brewer, J.A. (2015). Meditation leads to reduced default mode network activity beyond an active task. *Cognitive, Affective, & Behavioral Neuroscience*, 15, 712–720.

[15] Harris, C.R. et al. (2020). Array programming with NumPy. *Nature*, 585, 357–362.

[16] Harvey, A.C. (1989). *Forecasting, Structural Time Series Models and the Kalman Filter*. Cambridge University Press, Cambridge.

[17] Hasenkamp, W., Wilson-Mendenhall, C.D., Duncan, E. and Barsalou, L.W. (2012). Mind wandering and attention during focused meditation: a fine-grained temporal analysis of fluctuating cognitive states. *NeuroImage*, 59(1), 750–760.

[18] Jansen, M.J.W. (1999). Analysis of variance designs for model output. *Computer Physics Communications*, 117(1–2), 35–43.

[19] Kalman, R.E. (1960). A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1), 35–45.

[20] Kreutz, C. and Timmer, J. (2009). Systems biology: experimental design. *FEBS Journal*,

276(4), 923–942.

[21] Laukkonen, R.E. and Slagter, H.A. (2021). From many to (n)one: meditation and the plasticity of the predictive mind. *Neuroscience & Biobehavioral Reviews*, 128, 199–217.

[22] Ljung, L. (1999). *System Identification: Theory for the User* (2nd ed.). Prentice Hall, Upper Saddle River.

[23] Lutz, A., Slagter, H.A., Dunne, J.D. and Davidson, R.J. (2008). Attention regulation and monitoring in meditation. *Trends in Cognitive Sciences*, 12(4), 163–169.

[24] Oizumi, M., Albantakis, L. and Tononi, G. (2014). From the phenomenology to the mechanisms of consciousness: Integrated Information Theory 3.0. *PLoS Computational Biology*, 10(5), e1003588.

[25] Owen, A.B. (1998). Scrambling Sobol' and Niederreiter–Xing points. *Journal of Complexity*, 14(4), 466–489.

[26] Pukelsheim, F. (1993). *Optimal Design of Experiments*. Wiley, New York.

[27] Raichle, M.E., MacLeod, A.M., Snyder, A.Z., Powers, W.J., Gusnard, D.A. and Shulman, G.L. (2001). A default mode of brain function. *Proceedings of the National Academy of Sciences*, 98(2), 676–682.

[28] Rao, C.R. (1945). Information and the accuracy attainable in the estimation of statistical parameters. *Bulletin of the Calcutta Mathematical Society*, 37, 81–91.

[29] Raue, A., Kreutz, C., Maiwald, T., Bachmann, J., Schilling, M., Klingmüller, U. and Timmer, J. (2009). Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics*, 25(15), 1923–1929.

[30] Ros, T. et al. (2020). Consensus on the reporting and experimental design of clinical and cognitive-behavioural neurofeedback studies (CRED-nf checklist). *Brain*, 143(6), 1674–1685.

[31] Rothenberg, T.J. (1971). Identification in parametric models. *Econometrica*, 39(3), 577–591.

[32] Saltelli, A., Annoni, P., Azzini, I., Campolongo, F., Ratto, M. and Tarantola, S. (2010). Variance based sensitivity analysis of model output. Design and estimator for the total sensitivity index. *Computer Physics Communications*, 181(2), 259–270.

[33] Sandved-Smith, L., Hesp, C., Mattout, J., Friston, K., Lutz, A. and Ramstead, M.J.D. (2021). Towards a computational phenomenology of mental action: modelling meta-awareness and attentional control with deep parametric active inference. *Neuroscience of Consciousness*, 2021(1), niab018.

[34] Sitaram, R., Ros, T., Stoeckel, L., Haller, S., Scharnowski, F., Lewis-Peacock, J., Weiskopf, N., Blefari, M.L., Rana, M., Oblak, E., Birbaumer, N. and Sulzer, J. (2017). Closed-loop brain training: the science of neurofeedback. *Nature Reviews Neuroscience*, 18, 86–100.

[35] Sobol', I.M. (1993). Sensitivity estimates for nonlinear mathematical models. *Mathematical Modelling and Computational Experiments*, 1(4), 407–414.

[36] Tononi, G. (2004). An information integration theory of consciousness. *BMC Neuroscience*, 5, 42.

[37] Van Loan, C.F. (1978). Computing integrals involving the matrix exponential. *IEEE Transactions on Automatic Control*, 23(3), 395–404.

[38] Villaverde, A.F., Barreiro, A. and Papachristodoulou, A. (2016). Structural identifiability of dynamic systems biology models. *PLoS Computational Biology*, 12(10), e1005153.

[39] Virtanen, P. et al. (2020). SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17, 261–272.

[40] Vu, D. (2025). The brain as a query interface: thinking, reasoning, consciousness, and constructed reality. Unpublished manuscript, Phenikaa University, Hanoi; available from the author on request.

[41] Walter, E. and Pronzato, L. (1997). *Identification of Parametric Models from Experimental Data*. Springer, London.

[42] Weinstein, Y. (2018). Mind-wandering, how do I measure thee with probes? Let me count the ways. *Behavior Research Methods*, 50, 642–661.

[43] Wieland, F.-G., Hauber, A.L., Rosenblatt, M., Tönsing, C. and Timmer, J. (2021). On structural and practical identifiability. *Current Opinion in Systems Biology*, 25, 60–69.